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Research Article

Lifecycle updating of rockfill dam deformation models via dependence-aware sequential Bayesian inference with posterior-guided continual surrogate learning

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ABSTRACT

Rockfill dams exhibit pronounced stage-dependent deformation during construction, impoundment, and long-term operation, requiring monitoring-updated deformation models for lifecycle prediction and digital twin-based safety assessment. However, existing updating methods are often deterministic and stage-isolated, while conventional probabilistic approaches insufficiently capture dependence among mechanically related constitutive parameters. For multi-stage lifecycle analysis, these methods also lack cross-stage adaptability and often require repeated finite-element simulations or full surrogate retraining, leading to high computational cost. This study proposes a dependence-aware sequential probabilistic framework with posterior-guided continual surrogate learning for dynamic updating of rockfill dam deformation models. The framework progressively assimilates monitoring data to track lifecycle parameter evolution, preserves empirical parameter dependence via Copula-enhanced prior modeling within a sequential Bayesian updating scheme, and adaptively refines the surrogate by concentrating new training samples in posterior-relevant regions while retaining historical deformation knowledge. Application to the 303 m Lianghekou core-wall rockfill dam shows that the proposed method reduces the global relative settlement prediction error from 37.4% to 9.7% and reduces the dominant computational cost of lifecycle surrogate updating by 45.8% compared with full surrogate retraining. The updated model further enables uncertainty-informed settlement prediction, spatiotemporal deformation reconstruction, and posterior-informed diagnosis of safety-related deformation characteristics, providing an interpretable and efficient approach for lifecycle deformation analysis and digital twin-based dam safety management.

1. Introduction

Rockfill dams are widely used in large hydraulic projects because of their adaptability to complex geological conditions, use of locally available materials, and favorable seismic performance (Ma and Chi, 2016a; Zhang et al., 2026). As large granular structures, their deformation behavior evolves markedly during staged construction, reservoir impoundment, and long-term operation (Sun et al., 2026; Yang et al., 2025). Settlement is mainly governed by self-weight loading and compaction during construction, while wetting-induced deformation, hydraulic loading, stress redistribution, and creep become increasingly important after impoundment (Fig. 1). For tall rockfill dams, such lifecycle-dependent deformation is closely related to structural safety, seepage stability, and operational reliability, making reliable deformation assessment essential for dam safety management.

Field monitoring provides direct information on the in-situ deformation response of rockfill dams, but monitoring data are usually available only at discrete locations and cannot directly characterize the full-field mechanical state. Finite element (FE)-based deformation analysis provides a physics-based bridge between monitoring observations and continuous deformation fields, making monitoring-driven FE model updating an important approach for interpreting dam behavior,

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identifying material constitutive parameters, and supporting lifecycle safety assessment. Driven by advances in sensing, the Internet of Things, and artificial intelligence, civil infrastructure is evolving toward intelligent operation supported by digital twin (DT) technology (Nav et al., 2025; Saif et al., 2024). For rockfill dams, a practical DT is not merely a static replica or visualization tool, but an updatable computational model that can assimilate monitoring data, evolve with the physical dam, and support lifecycle performance assessment.

Existing monitoring-driven deformation model updating methods have improved dam behavior interpretation and deformation prediction, but several challenges remain for lifecycle applications. First, many back-analysis methods are deterministic and stage-specific, identifying a single optimal parameter set for a given period while providing limited information on parameter uncertainty and prediction reliability (Ai et al., 2024; Li et al., 2024). Because rockfill materials are inherently variable and monitoring data are limited and uncertain, uncertainty quantification is important for assessing model confidence and supporting risk-informed decisions (Chen et al., 2025; Dai et al., 2025). Second, conventional probabilistic updating often assumes independent prior parameters for simplicity (Li et al., 2024; Zeng et al., 2024), although rockfill constitutive parameters related to stiffness, volumetric deformation, strength, and nonlinear response are commonly correlated because of shared material properties, compaction conditions, stress states, and testing procedures (Ai et al., 2025; Yang et al., 2025). Neglecting such dependence may reduce the interpretability of posterior parameter evolution, especially when different parameter combinations can reproduce similar settlement responses (Huang et al., 2026). Third, lifecycle updating requires repeated model evaluations as new monitoring data become available and deformation mechanisms evolve. Direct FE-based Bayesian inference is computationally expensive, while conventional surrogate-assisted approaches are usually developed for a single analysis stage or fixed parameter domain (Yu et al., 2025; Zhang et al., 2026). A surrogate trained in earlier stages may lose accuracy in later stages, whereas full retraining requires additional FE samples and increases cumulative computational cost. Continual learning provides a potential solution by allowing the surrogate to incorporate new-stage information while retaining previously learned knowledge (Delange et al., 2021; Kirkpatrick et al., 2017; Van De Ven et al., 2022).

To address these issues, this study develops a dependence-aware sequential probabilistic framework with posterior-guided continual surrogate learning for lifecycle deformation model updating of rockfill dams. The proposed framework treats constitutive parameters as stage-wise effective material states and progressively updates their posterior distributions as monitoring data are assimilated. A Gaussian Copula-based dependence model is embedded into the posterior-to-prior propagation process, allowing posterior-informed marginal distributions from the previous lifecycle stage to be reconnected through an empirical dependence structure. To improve computational feasibility, a deep neural network surrogate is continually refined using posterior-guided FE samples, representative historical replay samples, and Fisher-information-weighted regularization. This strategy enables the surrogate to adapt to current-stage deformation characteristics while retaining historically learned response mappings.

The main contributions of this study are summarized as follows:

(1) A lifecycle-oriented sequential probabilistic updating formulation is developed for rockfill dam deformation models. By updating stage-wise effective constitutive parameter distributions rather than a single deterministic parameter set, the formulation enables progressive monitoring-data assimilation and uncertainty-informed deformation assessment across construction, impoundment, and operation stages.
(2) A dependence-preserving posterior-to-prior propagation strategy is introduced for mechanically related rockfill parameters. By reconnecting posterior-informed marginal distributions through an empirical Gaussian Copula dependence structure, the proposed approach retains prescribed statistical dependence among constitutive parameters and improves the interpretability of posterior parameter evolution.
(3) A posterior-guided continual surrogate learning strategy is proposed for efficient cross-stage model updating. The strategy uses posterior information to guide incremental FE sampling and combines response-stratified replay with Fisher-weighted regularization to balance current-stage adaptability and historical knowledge retention.
(4) The proposed framework is validated using the 303 m Lianghekou core-wall rockfill dam. The case study evaluates posterior parameter evolution, independent settlement prediction, surrogate updating performance, uncertainty decomposition, and safety-oriented deformation diagnosis based on the posterior-updated FE model, demonstrating its capability for lifecycle performance assessment of tall rockfill dams.

The remainder of this paper is organized as follows. Section 2 reviews related studies on monitoring-driven deformation model updating and digital twin-enabled lifecycle assessment of rockfill dams. Section 3 presents the proposed sequential probabilistic updating framework and posterior-guided continual surrogate learning strategy. Section 4 introduces the Lianghekou dam case study and implementation details. Section 5 presents the validation, comparative analysis, and discussion. Section 6 concludes the paper.

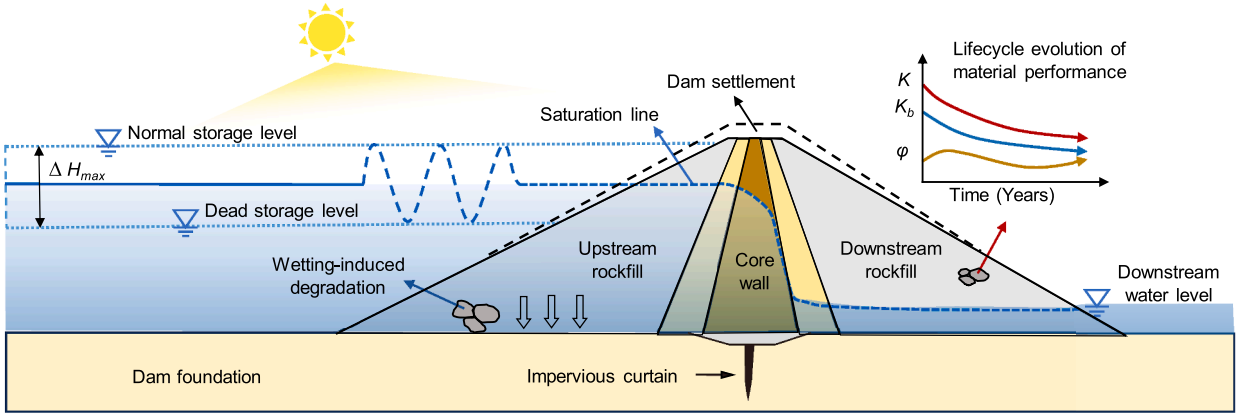


Fig. 1. Lifecycle deformation mechanisms and material evolution of rockfill dams.

2. Related works

2.1. SHM-oriented FE deformation model updating of rockfill dams

Deformation monitoring is a central component of structural health monitoring (SHM) for rockfill dams (Ma & Chi, 2016b), because settlement and displacement responses directly reflect staged construction, self-weight compaction, reservoir impoundment, wetting deformation, and long-term creep. However, monitoring data are usually available only at discrete locations and cannot directly describe the full-field mechanical state of the dam. Therefore, an important task in SHM-oriented safety assessment is to assimilate field observations into finite element-based (FE-based) deformation analysis models, so that monitored responses can be transformed into interpretable material parameters, deformation fields, and safety-related indicators.

FE-based deformation analysis provides the mechanistic basis for interpreting dam behavior (K. Wang et al., 2024; W. Zhou et al., 2011). By incorporating nonlinear constitutive models and simulating staged construction, reservoir loading, wetting, and creep, FE models can reproduce the main deformation characteristics of rockfill materials. Monitoring-assisted parameter back-analysis is widely used to reduce discrepancies between simulations and observations, while optimization-based methods improve parameter identification efficiency for deformation prediction and safety assessment (Jia & Chi, 2015; Z. Li et al., 2024).

Recent studies have enhanced monitoring-driven model updating through multi-source data integration, representative monitoring-point selection, response reconstruction, and surrogate-assisted inverse analysis, improving both reliability and computational efficiency (Ai et al., 2024, 2025; Huang et al., 2026). Meanwhile, probabilistic updating methods have been introduced to quantify parameter and prediction uncertainties by integrating prior information with monitoring data (Coletti et al., 2024; Ding et al., 2025; L. Zhang et al., 2018). Representative SHM-oriented deformation analysis and model updating studies are summarized in Table 1.

As summarized in Table 1, existing studies have advanced from FE-based mechanism interpretation to monitoring-driven model updating and uncertainty-informed deformation assessment. However, lifecycle SHM of rockfill dams still faces several limitations: many back-analysis and surrogate-assisted methods remain deterministic; probabilistic updating often neglects correlations among physically coupled constitutive parameters; and most surrogate-assisted strategies are developed for fixed domains or individual stages, whereas deformation mechanisms and credible parameter spaces evolve continuously from construction to long-term operation.

Table 1
Representative SHM-oriented deformation analysis and model updating studies for rockfill dams.

Category	Representative studies	Contribution
FE-based deformation analysis and deformation-mechanism interpretation	Zhou et al. (2011); Zhou et al. (2019); Wang et al. (2024)	Simulated settlement behavior, deformation mechanisms, and safety-related responses.
Optimization-based parameter back-analysis	Jia and Chi (2015); Ai et al. (2023); Li et al. (2024)	Calibrated constitutive parameters using monitoring data and optimization algorithms.
Multi-source monitoring and surrogate-assisted deterministic model updating	Ai et al. (2024, 2025); Ren et al. (2025); Huang et al. (2026)	Improved deterministic updating through multi-source data, reconstruction, and surrogate modeling.
Probabilistic model updating and uncertainty-informed deformation assessment	Zhang et al. (2018); Coletti et al. (2024); Ding et al. (2025)	Quantified parameter and prediction uncertainties through probabilistic calibration and monitoring-data assimilation.

2.2. Digital twin-enabled lifecycle performance assessment of rockfill dams

Digital twin technology is reshaping infrastructure management by integrating physical assets, monitoring data, computational models, and decision-support functions into an evolving cyber-physical system (Talasila et al., 2026). Recent studies show that infrastructure assessment is moving from static visualization toward multi-source data integration, dynamic state perception, uncertainty-aware prediction, and lifecycle decision support (Nakamura et al., 2025; X. Wang et al., 2025). Advances in physics-guided machine learning and sparse-sensor response reconstruction further support the inference of unobserved structural states from limited monitoring data (Ai et al., 2026).

This paradigm is particularly relevant to rockfill dams, whose deformation behavior evolves under staged construction, reservoir impoundment, wetting effects, stress redistribution, and long-term creep. Thus, a practical digital twin for rockfill dams should go beyond a virtual replica or monitoring dashboard, and should provide an updatable deformation analysis model for tracking material-state evolution, reconstructing spatial deformation characteristics, quantifying prediction uncertainty, and supporting lifecycle performance assessment. Recent studies have explored digital twin applications for deformation–seepage analysis, physics-constrained safety assessment, and full-field deformation reconstruction of high earth-rock dams (Ai et al., 2026; Huang et al., 2026; Tian et al., 2025). However, most existing studies remain oriented toward specific functions or representative stages. The continuous linkage among monitoring data assimilation, deformation model evolution, and lifecycle safety assessment has not yet been fully established, indicating the need for digital twin models that can evolve consistently with the dam state throughout construction, impoundment, and operation.

3. Methodology for sequential probabilistic lifecycle updating

3.1. Framework overview

To support lifecycle deformation analysis of rockfill dams, this study develops a sequential updating framework for probabilistically tracking material parameters and adaptively refining deformation analysis models. By progressively assimilating monitoring data from construction to impoundment and operation, the framework enables uncertainty-informed evaluation of deformation behavior and dam state under evolving conditions. The overall workflow is shown in Fig. 2.

The framework starts from dam geometry, material properties, environmental loads, and in situ deformation monitoring data. Based on the dam geometry and loading conditions, a high-fidelity finite element (FE) model is constructed to simulate the main deformation mechanisms of rockfill dams, including staged construction, reservoir impoundment, wetting deformation, and creep. By mapping constitutive parameters to deformation responses, the FE model serves as the physical forward model for assimilating monitoring data and updating the material constitutive parameters that govern deformation behavior, thereby supporting subsequent deformation assessment.

At each lifecycle stage, the effective constitutive parameters are updated through sequential probabilistic inference. The posterior distribution from the previous stage is propagated as informative prior knowledge, while newly acquired monitoring data further constrain the current parameter distributions. A dependence-informed prior construction is introduced to retain prescribed statistical dependence among mechanically related parameters, improving the interpretability

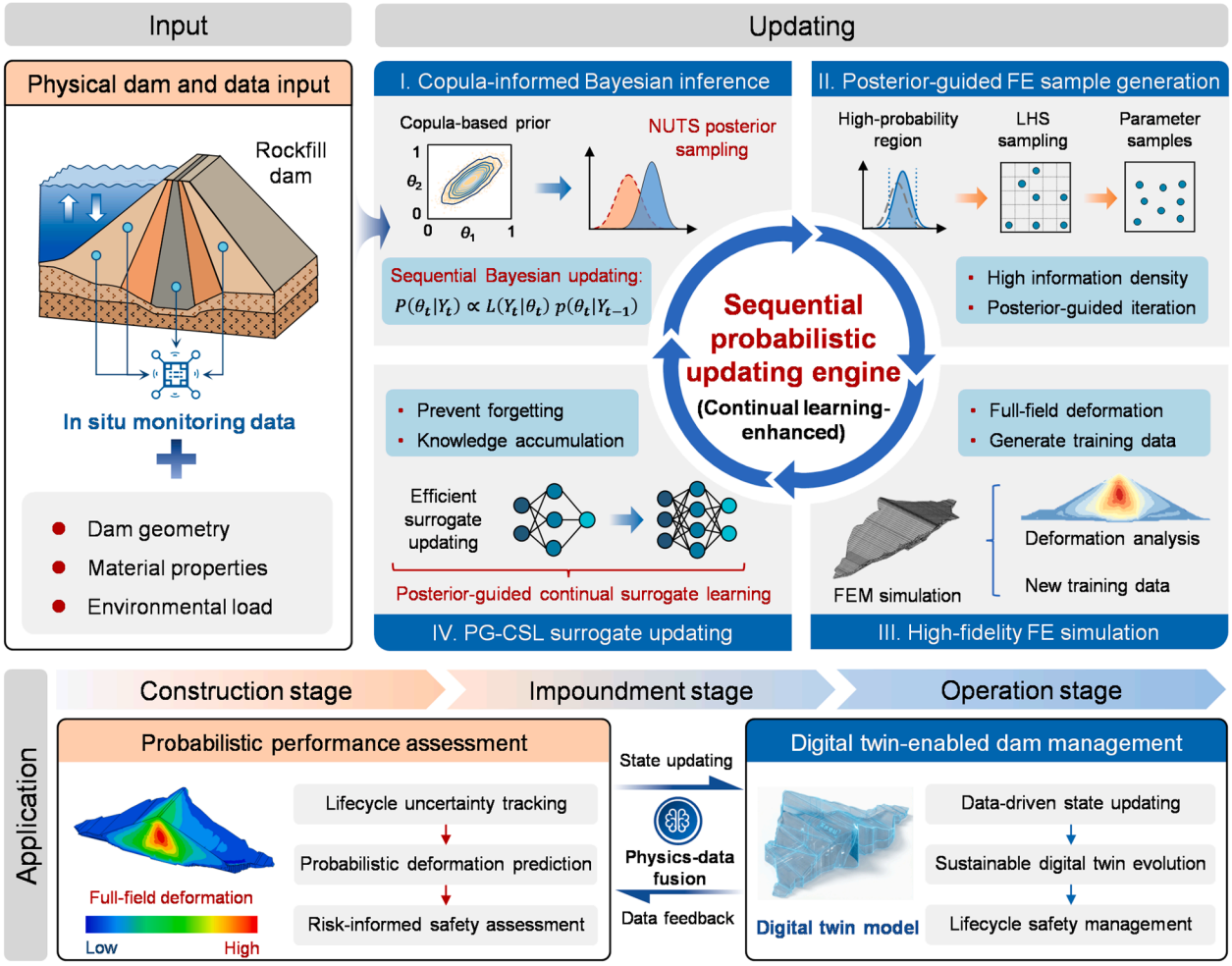


Fig. 2. Lifecycle sequential updating framework for digital twin-enabled rockfill dam deformation analysis.

of stage-wise parameter evolution. The resulting posterior distribution characterizes the current uncertainty of the effective material state and supports probabilistic deformation prediction.

The posterior distribution also guides surrogate evolution. New FE samples are generated within the posterior high-probability region, rather than repeatedly sampled from the original global parameter domain, and are evaluated by the high-fidelity FE model. Together with representative historical samples, these new samples are used to incrementally update the surrogate through the PG-CSL strategy. This allows the surrogate to focus on the parameter region most relevant to the current dam state while retaining previously learned deformation-response mappings, thereby reducing unnecessary FE simulations and avoiding full retraining at each stage.

Through this monitoring-driven process, the framework enables progressive uncertainty reduction, probabilistic deformation prediction, and posterior-informed deformation assessment over the dam lifecycle. The updated parameter distributions and reconstructed deformation responses provide a computational basis for digital twin-enabled dam management, including uncertainty tracking, deformation-state updating, and safety-oriented decision support.

3.2. Physics-based deformation forward model

The deformation behavior of rockfill dams is governed by material nonlinearity, staged construction, reservoir impoundment, wetting effects, and long-term creep. To represent these mechanisms within the proposed updating framework, a physics-based finite element (FE) deformation model is established as the forward model that maps constitutive parameters and loading conditions to deformation responses (Tomassini et al., 2025; W. Zhou et al., 2011).

The deformation of rockfill dams is generally composed of static, creep, and wetting deformation (Cheng et al., 2023; T. Zhou et al., 2025; W. Zhou et al., 2011), which act alternately or in combination during construction, impoundment, and long-term operation, thereby governing the overall deformation evolution of the dam. To represent these mechanisms, constitutive models commonly used in engineering practice are adopted to describe the nonlinear behavior of rockfill materials under different conditions. Specifically, static deformation is modeled by the Duncan–Chang E-B model (K. Wang et al., 2024), creep deformation by an exponential decay model (G. Li et al., 2004), and wetting deformation by a three-parameter wetting model (Jia et al., 2018). In this study, these three constitutive models are jointly employed as the governing equations of the deformation analysis model, and the associated parameters and their physical meanings are summarized in Table 2.

Within the sequential updating framework, the constitutive parameters of the deformation model are treated as stage-wise effective material parameters rather than fixed lifecycle constants. By assimilating field deformation monitoring data, the probabilistic updating process identifies parameter distributions that better represent the current mechanical state of the prototype dam. The updated parameter distributions quantify material-state uncertainty and provide the basis for posterior deformation prediction, full-field response reconstruction, and safety-oriented deformation assessment.

Table 2
Basic parameters and physical interpretation of the deformation analysis model.

Constitutive model	Parameter	Description
Duncan-Chang E-B model	K, K_b	Tangent modulus and bulk modulus coefficients
	n, m	Exponents for tangent and bulk moduli
	φ, $\Delta\varphi$	Friction angle and its reduction rate
	c	Cohesion of the material
	R_f	Failure ratio
Exponential decay empirical creep model	α	Rheological rate
	b, c, d	Creep model parameters
Three-parameter wetting deformation model	c_w, n_w, b_w	Wetting model parameters

3.3. Dependence-aware sequential probabilistic updating

3.3.1. Sequential Bayesian updating of constitutive parameters

For lifecycle deformation analysis of rockfill dams, the FE model establishes a physics-based mapping from constitutive parameters and loading conditions to deformation responses. However, the effective mechanical state of rockfill materials evolves with staged construction, reservoir impoundment, wetting deformation, stress redistribution, and long-term creep. Therefore, the parameters updated in this study are not regarded as fixed intrinsic constants over the entire lifecycle, but as stage-wise effective constitutive parameters that describe the apparent material response under the current engineering condition.

Let $D_t = \{y_{t,1}, y_{t,2}, \ldots, y_{t,n}\}$ denote the monitoring data newly assimilated at lifecycle stage t, and let $Y_{1:t} = \{D_1, D_2, \ldots, D_t\}$ denote the cumulative monitoring information up to this stage. The uncertain effective parameter vector at stage t is denoted by $\theta_t = (\theta_{t,1}, \theta_{t,2}, \ldots, \theta_{t,m})$, and the residual model-discrepancy standard deviation is denoted by $\sigma_{\mathrm{res},t}$. The surrogate-predicted deformation response is expressed as $F_{surr}(\theta_t)$.

For the i-th monitoring observation at stage t, the monitoring-prediction relationship is formulated as:

$$y_{i,t} = F_{\mathrm{surr},i}(\theta_t) + \varepsilon_{t,i} \tag{1}$$

Following Bayesian identification studies on the systematic treatment of uncertainty sources in model calibration (Rappel et al., 2019, 2020), the total response error $\varepsilon_{t,i}$ is decomposed as:

$$\varepsilon_{t,i} = \varepsilon_{\mathrm{obs},t,i} + \varepsilon_{\mathrm{surr},t,i} + \varepsilon_{\mathrm{disc},t,i} + \varepsilon_{\mathrm{res},t,i} \tag{2}$$

where $\varepsilon_{\mathrm{obs},t,i}$ denotes the observation error associated with the monitoring instrument, $\varepsilon_{\mathrm{surr},t,i}$ represents the surrogate approximation error, $\varepsilon_{\mathrm{disc},t,i}$ denotes the FE discretization error, and $\varepsilon_{\mathrm{res},t,i}$ represents the remaining response-level discrepancy caused by unresolved modeling assumptions, such as constitutive idealization, simplified boundary conditions, loading-history approximation, and material zoning simplification. The observation error is assigned according to the corresponding instrument type, the surrogate error is estimated from independent test-set performance following uncertainty-aware surrogate modeling studies (Deshpande et al., 2022), and the FE discretization error is specified using a quantity-of-interest-oriented conservative estimate, since the sequential updating primarily relies on settlement-related responses; following common engineering practice for numerical uncertainty characterization (Celik et al., 2008), this uncertainty is taken as approximately 1%–2% of the maximum lifecycle settlement response. The remaining discrepancy is treated as an unknown quantity and is estimated simultaneously with the constitutive parameters during the Bayesian updating process.

Assuming that the above error components are mutually independent and approximately Gaussian, the total variance of the i-th observation is expressed as

$$\sigma_{t,i}^2 = \sigma_{\mathrm{obs},t,i}^2 + \sigma_{\mathrm{surr},t,i}^2 + \sigma_{\mathrm{disc},t,i}^2 + \sigma_{\mathrm{res},t}^2 \tag{3}$$

Accordingly, the likelihood function at lifecycle stage t is defined as

$$L\left(D_t \middle| \theta_t, \sigma_{\mathrm{res},t}\right) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma_{t,i}} \exp\left\{ -\frac{\left[y_{t,i} - F_{\mathrm{surr},i}(\theta_t)\right]^2}{2\sigma_{t,i}^2} \right\} \tag{4}$$

This formulation separates explicitly quantifiable uncertainty sources from the remaining unexplained discrepancy, thereby avoiding the use of a single undifferentiated residual term. It also prevents surrogate and numerical errors from being implicitly attributed to constitutive parameters.

At stage t, the posterior distribution is written as:

$$p\left(\theta_t, \sigma_{\mathrm{res},t} \middle| Y_t\right) \propto L\left(D_t \middle| \theta_t, \sigma_{\mathrm{res},t}\right) \pi_t\left(\theta_t, \sigma_{\mathrm{res},t}\right) \tag{5}$$

where $\pi_t(\theta_t, \sigma_{\mathrm{res},t})$ is the prior distribution used at the current lifecycle stage. For the first stage, the prior is specified from laboratory tests, engineering experience, and the dependence-aware prior model described in Section 3.3.2. For subsequent stages, the current-stage prior is constructed from the posterior information obtained at the previous stage:

$$\pi_t\left(\theta_t, \sigma_{\mathrm{res},t}\right) \approx T\left[p\left(\theta_{t-1}, \sigma_{\mathrm{res},t-1} \middle| Y_{1:t-1}\right)\right] \tag{6}$$

where $T[\bullet]$ denotes the posterior-informed prior construction operator. In this study, this operator carries forward the posterior marginal distributions from stage t-1 and reconnects them through the Copula-based dependence structure to form the joint prior for stage t, while keeping the parameters within the physically admissible domain. Thus, the previous posterior is not used to enforce an unchanged parameter vector over the entire lifecycle; instead, it provides informative prior knowledge for updating the stage-wise effective parameter distribution when new monitoring data become available.

The resulting posterior distribution of θ_t characterizes the updated uncertainty of the effective constitutive parameters, whereas the posterior distribution of $\sigma_{\mathrm{res},t}$ quantifies the remaining response-level discrepancy after accounting for observation, surrogate, and discretization errors. The posterior samples are subsequently used for deformation prediction, uncertainty propagation, and posterior-guided surrogate updating.

Because the posterior distribution is high-dimensional, nonlinear, and affected by parameter dependence, it is sampled using the No-U-Turn Sampler (NUTS). NUTS is an adaptive variant of Hamiltonian Monte Carlo that automatically tunes the step size and trajectory length during sampling, thereby avoiding manual parameter adjustment and improving sampling efficiency compared with conventional random-walk-based MCMC methods (Coletti et al., 2024; Hoffman & Gelman, 2014). These features make it well suited for sequential Bayesian updating of multi-parameter deformation models. In this study, NUTS is used as an efficient posterior sampling tool, while the detailed sampling configuration and convergence diagnostics are provided in Section 4.4.2 and Appendix B.

3.3.2. Copula-enhanced prior modeling for parameter dependence

The prior construction operator $T[\bullet]$ in Eq. (6) determines how posterior information from one lifecycle stage is transferred to the next. For rockfill dam deformation updating, this operation should retain the monitoring-informed uncertainty learned from previous observations, while allowing the effective constitutive parameters to evolve with changing construction and operation conditions. It should also preserve the dependence among mechanically related parameters, because stiffness, strength, and nonlinear deformation parameters are often affected by similar material properties, compaction conditions, stress states, and laboratory testing procedures.

To this end, a dependence-preserving posterior-to-prior propagation

strategy is adopted. At lifecycle stage t >1, the posterior marginal distribution of each updated parameter from stage (t-1) is extracted and used as the corresponding prior marginal distribution for stage t:

$$F_{t,j}^{\text{prior}} = M_j[p(\theta_{t-1}|Y_{1:t-1})], \quad j = 1, 2, \ldots, m \tag{7}$$

where $F_{t,j}{}^{\text{prior}}$ denotes the prior cumulative distribution function of the j-th constitutive parameter at stage t, and M [•] denotes the marginalization operator for the j-th parameter. This operation propagates the marginal uncertainty constrained by earlier monitoring data, rather than transferring only posterior means or enforcing a time-invariant parameter vector over the full lifecycle.

After marginal propagation, mechanically related parameters are reconnected through a Gaussian-Copula dependence structure (Li et al., 2025; Yang et al., 2025). For a correlated parameter group g, a latent Gaussian vector is introduced as:

$$z_{t,g} \sim N(0, R_g) \tag{8}$$

where R_g is the empirical correlation matrix of group g. The latent variables are transformed into correlated uniform variables by:

$$u_{t,g,j} = \Phi(z_{t,g,j}) \tag{9}$$

and the physical parameters are obtained through inverse marginal transformation:

$$\theta_{t,g,j} = \left(F_{t,g,j}^{\text{prior}}\right)^{-1}(u_{t,g,j}) \tag{10}$$

where Φ [•] is the standard normal cumulative distribution function. This reparameterized formulation imposes dependence in the latent Gaussian space while allowing each constitutive parameter to retain its own marginal distribution in the physical parameter space. Therefore, different forms of marginal uncertainty representations can be consistently incorporated within the same prior-construction framework.

Equivalently, the joint prior density of correlated group g can be expressed as

$$\pi_{t,g}(\theta_{t,g}) = c_{R_g}\left[F_{t,g,1}^{\text{prior}}(\theta_{t,g,1}), \ldots, F_{t,g,d_g}^{\text{prior}}(\theta_{t,g,d_g})\right] \prod_{j=1}^{d_g} f_{t,g,j}^{\text{prior}}(\theta_{t,g,j}) \tag{11}$$

where d_g is the dimension of group g, $f_{t,g,j}{}^{\text{prior}}$ and $F_{t,g,j}{}^{\text{prior}}$ are the marginal probability density function and cumulative distribution function, respectively, and c_{Rg} [•] is the Gaussian Copula density governed by R_g. The complete prior distribution is then obtained by combining the Copula-based joint priors of all correlated parameter groups with the marginal priors of the remaining independently modeled parameters, forming the overall prior distribution of the full parameter vector at stage t.

For the initial lifecycle stage, the marginal priors are specified from laboratory tests and engineering experience. Bounded uniform distributions are adopted when the available information mainly provides physically admissible parameter ranges, whereas posterior-informed marginal distributions are used in subsequent stages after monitoring data have been assimilated. The correlation matrices R_g are estimated from an empirical database of laboratory triaxial test results for rockfill materials collected from 48 representative dams, covering concrete-faced and core-wall rockfill dams with different heights, lithologies, construction periods, and geographical regions. The project information is summarized inTable A.1 in Appendix A, and the detailed parameter data and statistical descriptions are provided by Huang et al. (2026). Considering the heterogeneity of the database and the monotonic but not necessarily linear relationships among constitutive parameters, Spearman correlation coefficients are used to quantify the empirical parameter correlations. The resulting correlations provide practical prior information for constructing the grouped Copula priors. As shown in Fig. 3, clear positive correlations are observed among several

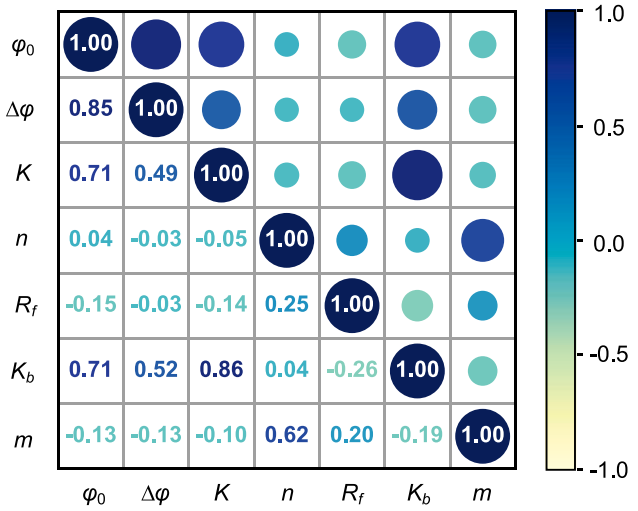


Fig. 3. Correlation coefficient matrix of constitutive parameters for rockfill materials.

mechanically related parameters, such as K and K_b, φ_0 and $\Delta\varphi$, providing practical prior information for constructing the grouped Copula priors.

In this study, the Gaussian Copula is not used as a standalone static prior, but as a correlation-reconnection mechanism in the sequential updating process. It reconnects evolving posterior-informed marginal distributions into a joint prior at each lifecycle stage, so that marginal uncertainty can be progressively updated by field monitoring data while prescribed empirical correlations among mechanically related parameters are retained. This strategy reduces the tendency of correlated parameters to evolve in a decoupled manner because of compensation effects in displacement-based back analysis, and leads to more stable and physically interpretable parameter evolution during sequential updating.

The correlation matrices derived from the 48-dam database are used as empirical prior information rather than exact project-specific material laws for the LHK dam. Project-specific deformation observations are subsequently assimilated through the likelihood function at each lifecycle stage, allowing the posterior distributions to be updated by field monitoring data. In addition, the Gaussian Copula primarily captures monotonic dependence through correlation matrices and may have limited capability in representing tail dependence or more complex nonlinear dependence structures. More flexible models, such as t-Copulas or vine Copulas, could potentially provide richer descriptions under extreme safety states, although their reliable identification typically requires larger databases or more project-specific observations. This issue will be investigated in future work.

The posterior samples generated from this sequential inference are further used for uncertainty propagation and for defining the posterior high-probability region that guides incremental FE sampling in the continual surrogate learning strategy described in Section 3.4.

3.4. Posterior-guided continual surrogate learning strategy

3.4.1. DNN surrogate modeling and Continual learning strategy

In sequential probabilistic updating of rockfill dam deformation models, repeated evaluations of the high-fidelity FE model are required for likelihood calculation, posterior sampling, and uncertainty propagation. Directly embedding the nonlinear three-dimensional FE model into Bayesian inference would lead to prohibitive computational cost (Ai et al., 2024; Yu et al., 2025). Therefore, a DNN-based surrogate model is introduced to approximate the mapping from uncertain constitutive parameters and spatiotemporal loading information to deformation responses.

The surrogate is formulated as a multi-output regression model

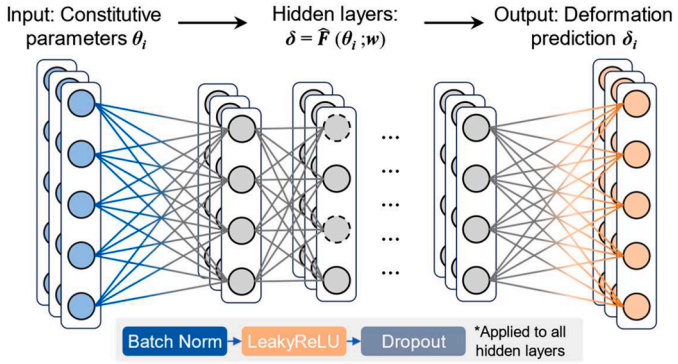


Fig. 4. Architecture of the deep neural network surrogate model.

(Fig. 4). Its inputs include the updated constitutive parameters and the corresponding spatiotemporal features of the monitoring or prediction locations, while its outputs are the deformation responses at representative monitoring points or structural locations. The network adopts a fully connected feedforward architecture with batch normalization, LeakyReLU activation, and dropout regularization, which improves training stability and generalization under limited FE samples. Once trained, the surrogate replaces the FE model during Bayesian sampling and enables efficient posterior inference.

However, a static surrogate trained only once is insufficient for lifecycle deformation analysis. During construction, reservoir impoundment, and long-term operation, the relevant parameter domain and deformation response pattern change as the dominant mechanisms evolve from self-weight compaction to wetting-induced deformation and creep. A surrogate trained only on earlier-stage samples may lose accuracy in later stages, whereas full retraining at every lifecycle stage requires additional FE simulations and increases cumulative computational cost.

To address this issue, the surrogate is updated in a continual-learning manner. Continual learning provides a feasible strategy for sequentially incorporating new information while preserving previously learned knowledge, aiming to balance model plasticity and stability (Delange et al., 2021; Parisi et al., 2019). In the context of lifecycle dam analysis, this enables the surrogate to continuously adapt to newly generated FE samples and evolving deformation patterns without losing predictive capability for earlier stages. Specifically, the surrogate trained at the previous lifecycle stage is inherited as the initial model for the current stage and is incrementally updated to absorb newly available sample information while maintaining consistency with previously learned response characteristics. This strategy allows the surrogate to adapt to current-stage deformation features while retaining previously learned response mappings. The detailed implementation of this lifecycle-adaptive updating strategy is described in Section 3.4.2.

3.4.2. *Posterior-guided continual surrogate updating*

In the proposed framework, the posterior distribution obtained from Bayesian updating is used not only for uncertainty quantification, but also as an adaptive learning signal for surrogate evolution. Instead of generating new FE samples from the original global parameter bounds, the incremental sampling domain is defined by the posterior high-probability region of the previous lifecycle stage. In this study, this region is taken as the 90% credible interval of each updated parameter and is restricted within the physically admissible parameter space. Latin hypercube sampling (LHS) is then performed within this posterior-defined domain to maintain space-filling coverage and avoid excessive concentration around posterior modes. In this way, new FE simulations are concentrated in the parameter region most relevant to the current dam state, enabling local refinement of the surrogate with fewer additional FE samples and reducing extrapolation risk during lifecycle transitions.

At lifecycle stage t, the incremental training dataset is constructed by combining newly generated FE samples and representative replay samples:

$$D_t^{\mathrm{inc}} = D_t^{\mathrm{new}} \cup D_t^{\mathrm{rep}} \tag{12}$$

where D_t^{new} denotes the new FE samples generated from the posterior-defined credible domain, and D_t^{rep} denotes historical replay samples selected from the sample repository. The new samples enable the surrogate to learn current-stage deformation features associated with updated loading conditions, reservoir level, wetting effects, and creep evolution, whereas the replay samples preserve historically validated response mappings and mitigate catastrophic forgetting. To retain representative historical information, the replay repository is constructed using response-stratified sampling, in which samples are selected from different deformation-response intervals.

The surrogate model at stage t is initialized using the optimized weights from stage t-1, thereby inheriting the previously learned mapping between constitutive parameters and deformation responses. This weight transfer provides a stable starting point for incremental learning and reduces the need for extensive retraining. During incremental training, the model is updated using the combined dataset D_t^{inc}, which integrates newly generated samples reflecting current-stage conditions and replay samples preserving historical knowledge. To balance adaptation and retention, the objective function combines the regression loss on D_t^{inc} with a Fisher-information-weighted regularization term:

$$L_t(w) = L_{\mathrm{inc}}\left(w; D_t^{\mathrm{inc}}\right) + \frac{\lambda}{2}\sum_i F_i\left(w_i - w_{i,t-1}^*\right)^2 \tag{13}$$

where w denotes the current network weights, $w_{i,t\text{-}1}^*$ is the optimized value of the i-th weight from the previous stage, F_i is the Fisher-information-based importance of w_i, and λ controls the strength of historical-knowledge retention.

In this formulation, the Fisher information serves as a quantitative measure of parameter sensitivity with respect to previously learned tasks, thereby guiding the selective preservation of critical knowledge during incremental updating. Weights with larger F_i values are constrained more strongly to preserve previously learned response mappings, whereas less important weights remain more flexible for adapting to new deformation features, enabling a balanced trade-off between stability and adaptability across lifecycle stages.

As illustrated in Fig. 5, the proposed posterior-guided continual surrogate learning (PG-CSL) strategy consists of three coupled modules: posterior-driven sampling, historical information replay, and Fisher-weighted incremental learning. The posterior distribution determines where new FE information should be added, response-stratified replay determines which historical samples should be retained, and Fisher-weighted regularization controls how the network should be updated. This enables the surrogate to continuously adapt across construction, impoundment, and operation stages while maintaining historical prediction consistency with reduced FE-sample generation cost.

4. Project case

4.1. *Project Overview*

The proposed framework is validated using the Lianghekou (LHK) tall core-wall rockfill dam on the Yalong River in Sichuan Province, China. The dam has a maximum height of 303 m, a crest elevation of 2878 m, and a crest width of 16 m. The normal and dead water levels are 2865 m and 2785 m, respectively, with a total reservoir capacity of approximately 10.767×10^9 m^3. As shown in Fig. 6(c), the dam adopts a typical zoned configuration consisting of an impervious core, filter layers, transition zones, and rockfill zones.

Owing to its large height, staged construction process, and long-term

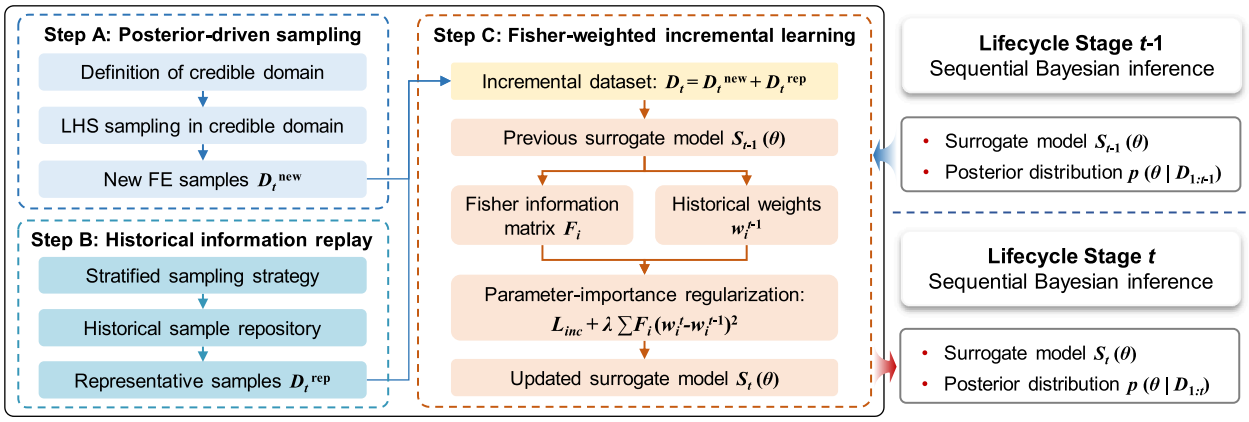


Fig. 5. Posterior-guided continual surrogate learning across lifecycle stages.

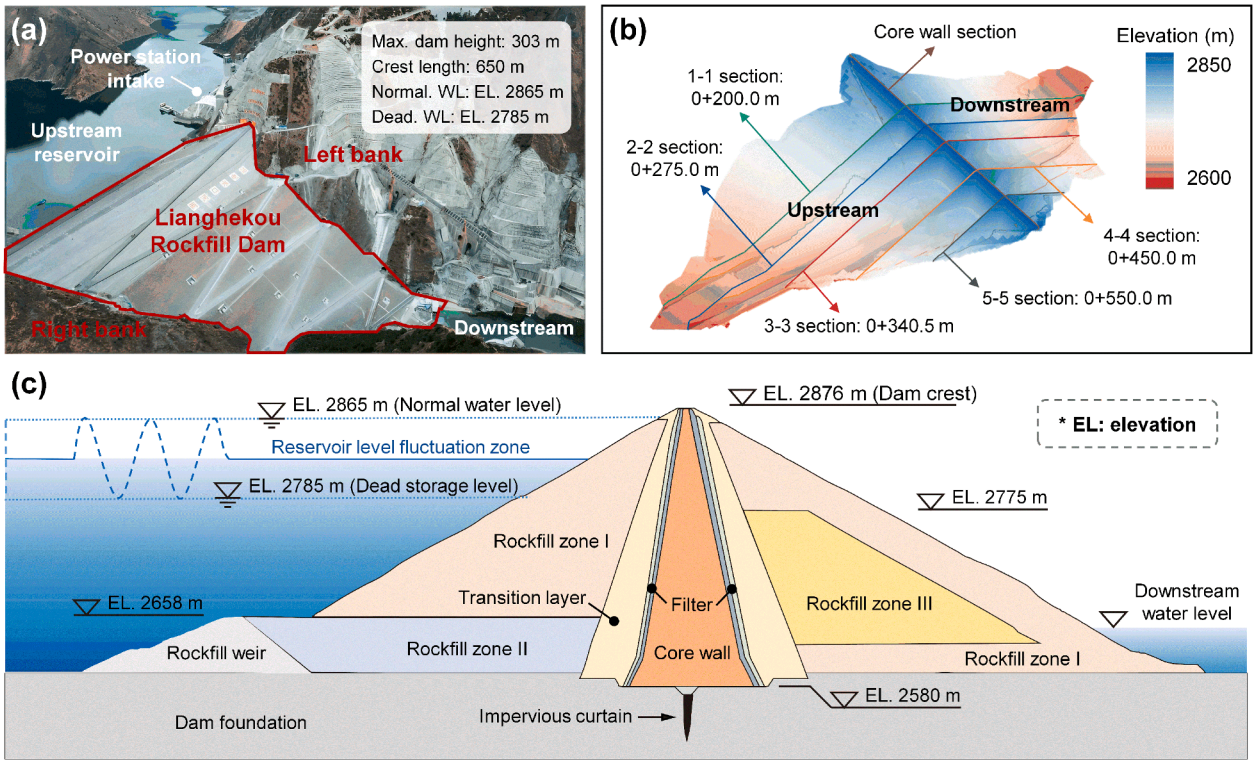


Fig. 6. Overview of the Lianghekou project: (a) Aerial view of the rockfill dam; (b) 3D dam geometry with monitoring sections; (c) Typical cross-section and material zoning of the dam.

exposure to reservoir fluctuations, the LHK dam exhibits pronounced lifecycle-dependent deformation during construction, impoundment, and operation. Its deformation behavior is governed by the combined effects of static compaction, wetting-induced deformation, and creep, making it a representative case for investigating deformation evolution and uncertainty propagation in rockfill dams. A comprehensive monitoring system has been established across several representative cross-sections covering the key stress-bearing regions of the dam (Fig. 6 (b)). These sections include multiple instruments for measuring internal settlement, horizontal displacement, and core-wall deformation, providing reliable field data for deformation analysis, sequential Bayesian updating, and uncertainty quantification.

4.2. Finite element modeling and physics-based simulation

A three-dimensional finite element (FE) model was developed in ABAQUS to simulate the stress-deformation evolution of the LHK core-wall rockfill dam during construction and operation. A user-defined material subroutine (UMAT) was implemented to capture the nonlinear constitutive behavior of rockfill materials. The model explicitly represents the dam zoning, including the impervious core, filter, transition, and rockfill zones. The computational domain consists of 150,776 elements and 146,691 nodes, primarily discretized using eight-node hexahedral elements. Considering the nearly 300 m dam height, complex zoned configuration, and representative monitoring sections used for deformation updating, the adopted mesh resolution is consistent with the discretization level commonly used in large-scale three-dimensional analyses of rockfill dams (Ai et al., 2025; Ren et al., 2025; Yu et al., 2025). The material zoning and FE mesh are shown in Fig. 6(c) and Fig. 7(a), respectively.

Boundary conditions were specified according to site constraints. The foundation base was fully fixed to represent competent bedrock,

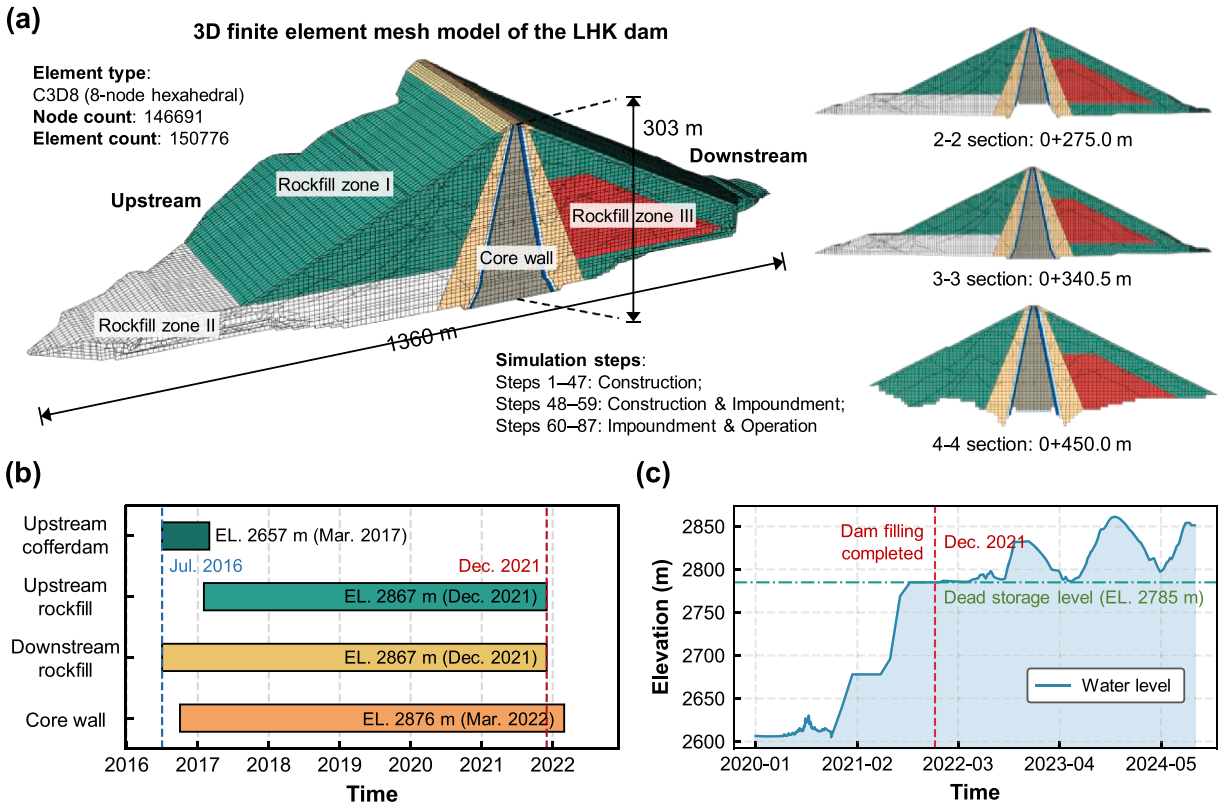


Fig. 7. Configuration of the FE-based simulation model: (a) FE mesh and representative sections; (b) Dam filling sequence; (c) Reservoir impoundment process.

while normal displacements along the lateral boundaries were restrained to account for topographic confinement. The loading process followed the documented construction and impoundment schedule. Dam filling began in July 2016 and was completed in December 2021. Initial reservoir impoundment started in October 2020, and normal operation commenced in July 2023. Accordingly, 87 analysis steps were defined, including 59 construction steps for layer-by-layer filling and subsequent steps for staged impoundment and operational conditions.

The FE model incorporates the static, creep, and wetting-induced deformation components described in Section 3.2. The initial constitutive parameters were derived from large-scale triaxial tests, as listed in Table A.2, and were used to provide a physically informed basis for defining the initial prior parameter ranges in the Bayesian updating process.

4.3. Lifecycle staging and monitoring data partitioning

To implement sequential probabilistic updating, the continuous dam construction–operation process was divided into four representative lifecycle stages according to the documented construction and impoundment schedule of the LHK dam. Phase I (December 2019) corresponds to a filling elevation of 2755 m, approximately three-quarters of the dam height, when deformation was mainly governed by self-weight loading and static compaction. Phase II (December 2021) represents dam completion, with deformation still dominated by static response. Phase III (July 2023) corresponds to the initial operation period, during which reservoir impoundment induced wetting deformation in the upstream rockfill. Phase IV (October 2024) represents the first attainment of the normal pool level, when sustained high-water exposure led to coupled wetting and creep deformation.

High-quality in situ monitoring data are essential for model updating and uncertainty quantification (G. Ma et al., 2024). The monitoring dataset used in this study consists of multi-source in situ deformation measurements collected from embedded instruments distributed across the main rockfill zones and the core wall. Before Bayesian updating, a monitoring-data filtering procedure was applied to remove low-quality or redundant records and retain measurements with higher information content for model updating (Ai et al., 2024). Candidate monitoring points were evaluated using quality and representativeness indicators, and the selected monitoring points, together with their instrument-specific nominal precisions used in the likelihood formulation, are summarized in Table 3.

To provide an independent benchmark for posterior predictive evaluation, 10 monitoring points with complete lifecycle records were reserved as validation data. These records were excluded from likelihood construction, parameter updating, and model calibration, and were used only for subsequent comparative validation. The remaining monitoring points were used for sequential Bayesian calibration.

Table 3
Distribution of selected monitoring points and nominal instrument precisions.

Material partition	Number of points	Monitoring instruments			
		ESG (± 3mm)	SAA (± 0.5mm)	HOSG (± 3mm)	SDT (± 0.01mm)
Rockfill zone I	13	3	2	4	4
Rockfill zone II	7	4	0	2	1
Rockfill zone III	12	5	0	7	0
Core wall	16	6	8	0	2
Total	48	18	10	13	7

Note: ESG (electromagnetic settlement gauge), SAA (shape-accel-array), HOSG (hydraulic overflow settlement gauge), and SDT (soil displacement transducer). The values in parentheses denote the nominal measurement precision of each instrument.

4.4. Configuration of the sequential updating framework

4.4.1. Surrogate model and continual learning settings

A DNN-based surrogate was constructed to replace repeated high-fidelity FE evaluations during sequential Bayesian inference. As shown in Fig. 4, the surrogate approximates the nonlinear mapping from constitutive parameters to deformation responses at key monitoring points across representative dam sections. For the four main material zones of the LHK dam, the candidate uncertain parameters were selected according to the deformation model in Section 3.2 and prior sensitivity analyses (Y. Li et al., 2013), including φ_0, $\Delta\varphi$, K, n, K_b, m, α, b, c_{creep}, d, c_w, n_w, and b_w.

To reduce dimensionality while retaining the structure of the experimentally calibrated parameter groups (Ai et al., 2025; Miao et al., 2025), two multiplicative scaling factors, λ_{creep} and $\lambda_{wetting}$, were introduced. The factor λ_{creep} jointly scales the creep-related parameters (b, c_{creep}, d) in zones where creep deformation is considered, whereas $\lambda_{wetting}$ scales the wetting-related parameters (c_w, n_w, b_w) only in rockfill zones subjected to wetting. The final updating vector therefore consisted of nine parameters: φ_0, $\Delta\varphi$, K, n, K_b, m, α, λ_{creep} and $\lambda_{wetting}$. Their prior ranges were determined from large-scale triaxial tests and engineering experience (Table A.3) and specified as broad physically admissible intervals to support global initial sampling.

In Phase I, 300 parameter sets were generated by Latin hypercube sampling (LHS) and evaluated using the FE model to train the base surrogate (Miao et al., 2025). The DNN adopted a fully connected feedforward architecture with three hidden layers of 256, 192, and 128 neurons, as determined by Bayesian hyperparameter optimization. This base model was used for Phase I Bayesian updating and inherited as the initial surrogate for later lifecycle stages. In Phases II–IV, the surrogate was updated incrementally using posterior-guided FE samples and historical replay samples instead of being retrained from scratch.

The dropout rates, EWC penalty coefficient, and replay ratio were selected through preliminary sensitivity tests and validation-based screening to balance current-stage adaptability and historical-stage retention. The replay buffer was constructed from the historical FE sample repository using response-stratified sampling, where the response range was divided into 25 quantile bins to preserve diverse deformation patterns. A replay ratio of 0.5 was used in Phases II-IV, meaning that the number of replay samples was approximately 50% of the newly generated FE samples at each incremental stage. The final surrogate architecture and continual learning settings are summarized in Table 4.

4.4.2. Implementation of sequential Bayesian updating

The sequential Bayesian updating framework was implemented in Pyro, with posterior inference performed using the No-U-Turn Sampler (NUTS). At each lifecycle stage, the uncertain constitutive parameters and residual model discrepancy were jointly sampled. The trained surrogate model was used for likelihood evaluation to avoid repeated direct FE simulations during MCMC sampling, and automatic differentiation was employed to compute the gradients required by NUTS.

The likelihood variance incorporated four uncertainty components: observation error, surrogate approximation error, FE discretization error, and residual model discrepancy. The observation error was assigned according to the nominal precision of each monitoring instrument, while the surrogate error was estimated from the independent test-set performance of the surrogate model at each lifecycle stage. The FE discretization error was specified as 50 mm, corresponding to a conservative representative value of approximately 1%-2% of the maximum lifecycle settlement response. The residual discrepancy was treated as an unknown parameter and inferred jointly with the constitutive parameters.

For each lifecycle stage, four independent Markov chains were run in parallel. Each chain included 1000 warm-up steps and 2000 posterior sampling steps. The target acceptance probability was set to 0.8, and the maximum tree depth was set to 8. Sampling convergence was assessed using the Gelman–Rubin statistic and effective sample size. The main NUTS-MCMC settings and error-term definitions are summarized in Table 5.

Table 4
Configuration of the surrogate model architecture and continual learning strategy.

Category	Item	Specification / Value
Network architecture	Model type	Fully-connected feedforward neural network
	Hidden layers	3 layers (Neurons: 256, 192, 128)
	Activation function	LeakyReLU ($\alpha = 0.01$)
	Regularization	Batch Normalization + Dropout (0.2, 0.15, 0.1)
Baseline training (Phase I)	Data splitting	70% Training / 20% Validation / 10% Testing
	Optimizer	Adam (Weight decay $= 1 \times 10^{-4}$)
	Loss function	Mean Squared Error (MSE)
	Learning rate	1×10^{-3} with Cosine Annealing schedule
	Training epochs	200
Continual Learning (Phase II–IV)	Algorithm	Posterior-guided continual surrogate learning
	EWC penalty (λ)	2000
	Fisher information	Diagonal approximation
	Replay ratio	0.5
	Replay sampling	Stratified sampling (25 quantile bins)
	Incremental epochs	200

4.5. Lifecycle probabilistic model updating and deformation assessment

4.5.1. Surrogate updating and posterior parameter inference

The reliability of the proposed updating framework was first evaluated in terms of surrogate prediction accuracy, MCMC convergence, posterior parameter evolution, and uncertainty-source decomposition. These results provide the basis for subsequent lifecycle deformation prediction and safety-oriented assessment.

Table 6 summarizes the predictive performance of the DNN surrogate across the four lifecycle phases using independent test samples that were excluded from training, validation, and sequential updating. The surrogate maintained high accuracy during lifecycle updating, with R^2 increasing from 0.963 in Phase I to 0.988 in Phase IV, while RMSE and MAE decreased from 8.3 cm and 6.4 cm to 5.6 cm and 4.2 cm,

Table 5
NUTS-MCMC settings and likelihood error-term definitions.

Category	Item	Specification / Value
Algorithm core	Probabilistic framework	Pyro
	Sampling method	No-U-Turn Sampler (NUTS)
	Gradient Calculation	Automatic Differentiation (Autograd)
Chain settings	Number of chains	4
	Warm-up (Burn-in) steps	1000 per chain
	Sampling steps	2000 per chain
Hyperparameters	Target acceptance probability	0.8
	Max tree depth	8
Convergence diagnostics	Gelman-Rubin statistic	< 1.05
	Effective sample size (ESS)	> 1000
Error-term settings	Observation error σ_{obs}	Instrument accuracy (see Table 3)
	Surrogate error σ_{surr}	Independent validation
	FE discretization error σ_{disc}	50 mm
	Residual discrepancy σ_{res}	Jointly inferred by Bayesian inference

Table 6
Performance metrics of the surrogate model.

Phase	Description	Incremental samples	Cumulative samples	R^2	RMSE (cm)	MAE (cm)
Phase I	Construction	300	300	0.963	8.3	6.4
Phase II	Construction	150	450	0.970	8.2	6.3
Phase III	Initial Impoundment	100	550	0.984	6.1	4.5
Phase IV	Operation	100	650	0.988	5.6	4.2

respectively. As shown in Fig. 8, the surrogate predictions agree well with FE simulation results across all phases. Meanwhile, the number of newly generated FE samples decreased from 300 in Phase I to 100 in Phases III-IV, indicating that posterior-guided incremental learning improves local surrogate accuracy while reducing additional FE simulations.

The convergence of Bayesian sampling was assessed using representative trace plots and posterior density distributions of the NUTS chains. As shown in Fig. B.1, the four parallel chains fluctuate around stable levels without evident non-stationary drift, and the posterior density estimates from different chains show strong overlap. The quantitative diagnostics also satisfy the prescribed criteria, with the Gelman-Rubin statistic below 1.05 and effective sample size exceeding 1000. These results indicate satisfactory chain mixing and support the reliability of the posterior samples used for parameter interpretation and uncertainty propagation.

Figure 9 presents the marginal posterior distributions of representative parameters across the four lifecycle phases, where the remaining dimensions are implicitly marginalized through MCMC sampling. The corresponding posterior means and standard deviations are summarized in Table 7. Overall, the posterior distributions become more concentrated as monitoring data are progressively assimilated, indicating reduced uncertainty in the identified effective material parameters. Mechanically related parameters within the same material zone, such as *K*, *Kb*, and φ_0, exhibit coordinated evolutionary trends, suggesting that the Copula-enhanced prior construction helps preserve prescribed parameter dependence and improves the interpretability of posterior parameter evolution.

The uncertainty components in the likelihood function were further decomposed to examine their lifecycle evolution. As shown in Fig. 10, FE discretization error, surrogate approximation error, and residual discrepancy are the main uncertainty sources, whereas observation error remains negligible because of the high precision of the monitoring instruments. The contribution of surrogate error decreases from 54.5% in Phase I to 26.3% in Phase IV, demonstrating the effectiveness of sequential surrogate updating. The residual discrepancy σ_{res}, which is

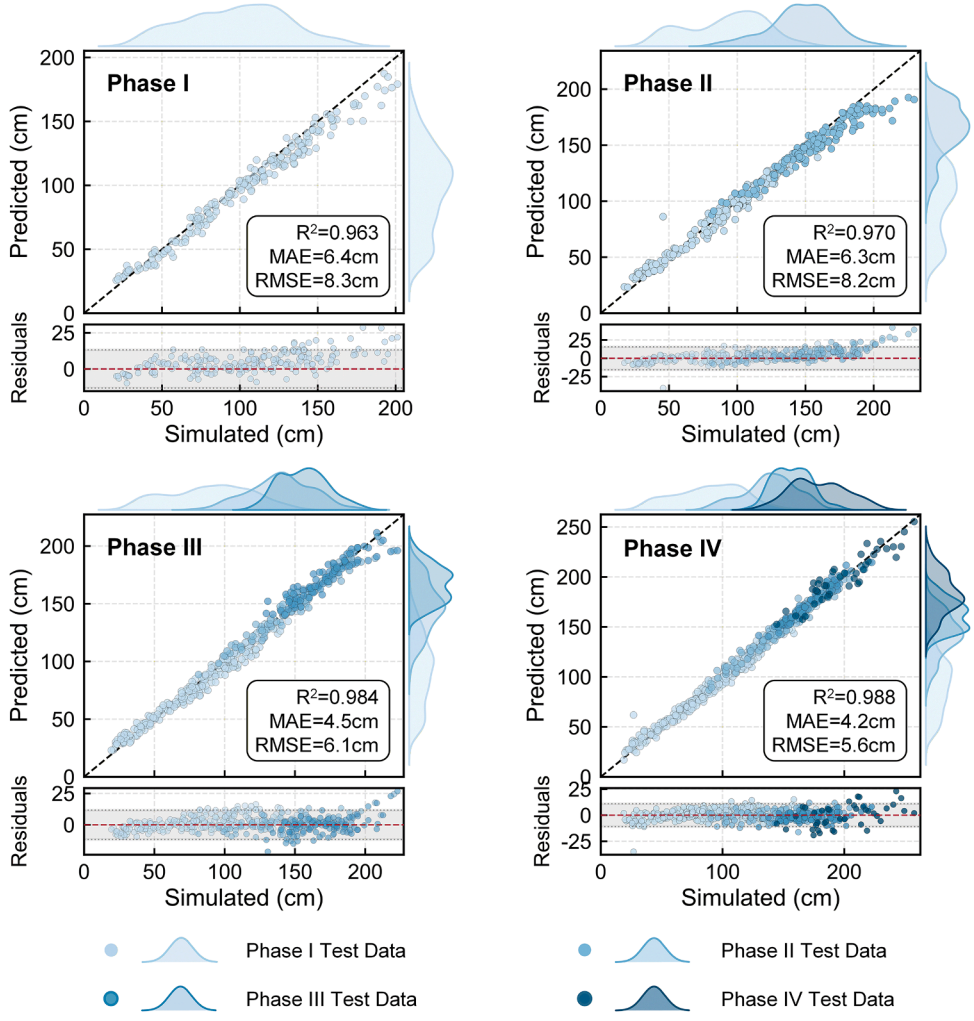


Fig. 8. Predictive performance of the continual learning surrogate model across lifecycle phases.

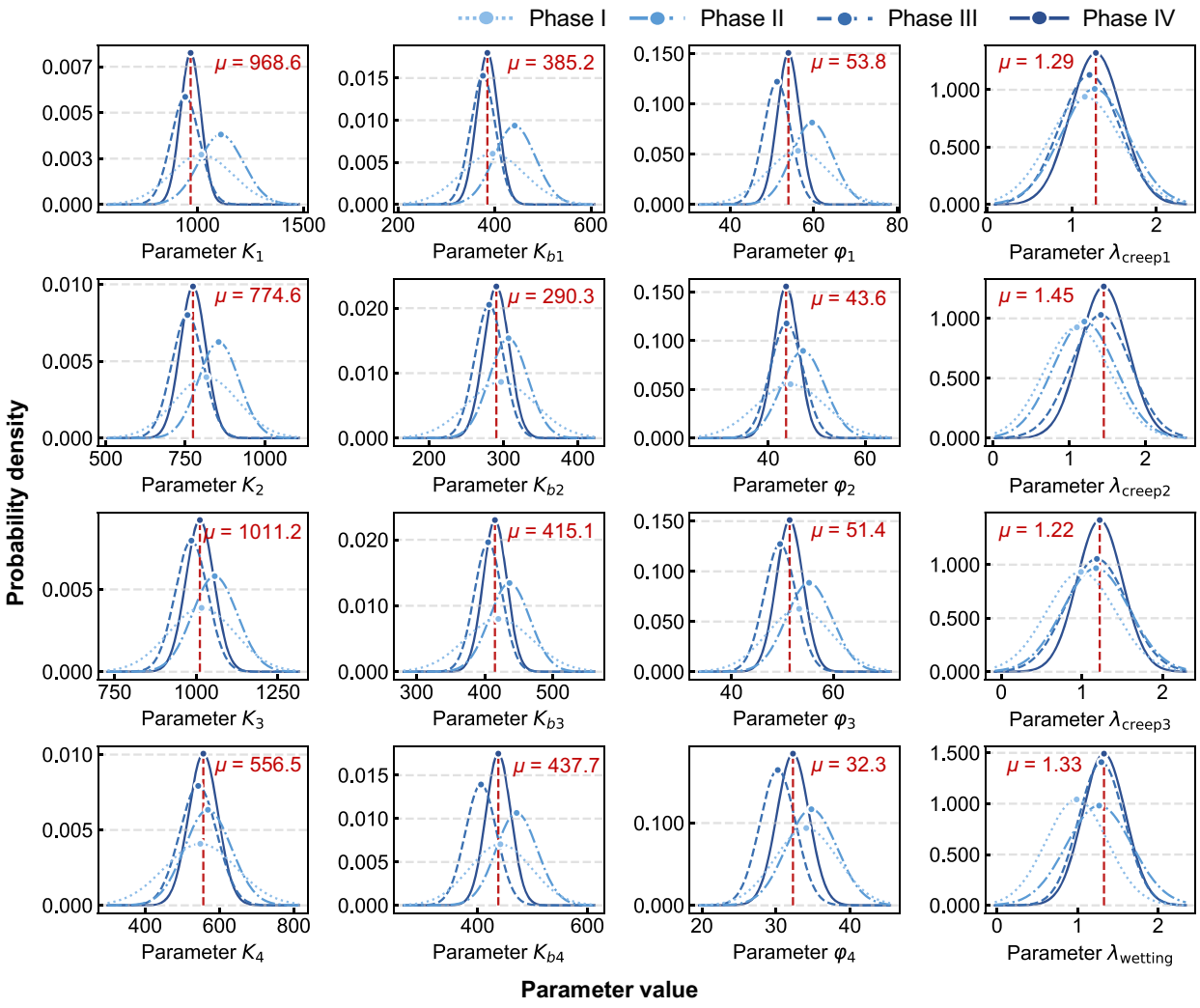


Fig. 9. Evolution of marginal posterior probability densities of key material parameters across lifecycle phases.

Table 7
Posterior statistics of representative parameters and uncertainty terms across lifecycle stages.

Item	Parameter	Statistic	Phase I	Phase II	Phase III	Phase IV
Rockfill zone I	K	Mean	1019.06	1111.54	943.19	968.64
		Standard deviation	146.65	104.63	68.07	48.35
	K_b	Mean	396.16	441.84	376.07	385.22
		Standard deviation	66.12	42.62	26.13	22.19
	φ_0	Mean	56.26	59.63	51.23	53.95
		Standard deviation	7.46	4.89	3.27	2.65
	λ_{creep}	Mean	1.16	1.27	1.21	1.29
		Standard deviation	0.42	0.40	0.35	0.30
	λ_{wetting}	Mean	0.99	1.27	1.30	1.33
		Standard deviation	0.38	0.41	0.28	0.27
Core wall	K	Mean	548.43	568.87	542.75	556.48
		Standard deviation	97.68	62.95	50.35	39.65
	K_b	Mean	441.52	471.09	406.20	437.72
		Standard deviation	56.61	37.42	28.63	22.81
	φ_0	Mean	34.09	34.78	30.25	32.30
		Standard deviation	4.25	3.42	2.43	2.16
	λ_{creep}	Mean	0.72	0.74	0.84	1.06
		Standard deviation	0.39	0.25	0.23	0.20
Residual discrepancy	σ_{res}	Mean	56.90	71.25	85.73	79.34
		Standard deviation	7.90	7.48	7.40	7.34

inferred jointly with the constitutive parameters, represents the remaining response-level mismatch after accounting for observation, surrogate, and discretization errors. It exhibits a clear lifecycle-dependent trend: it is relatively small during construction, increases

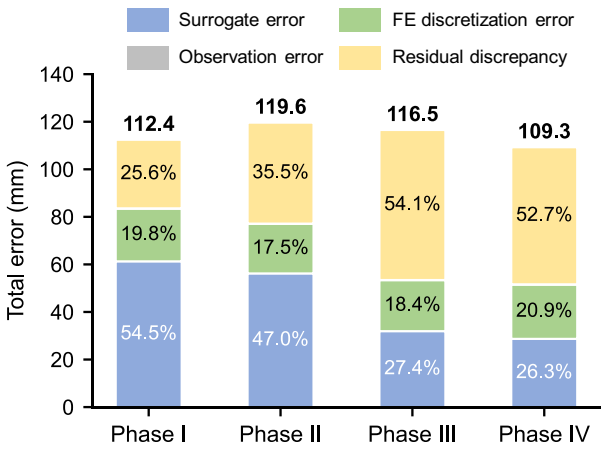


Fig. 10. Lifecycle evolution of uncertainty contributions.

during impoundment and early operation, and decreases moderately in the later operation stage. This behavior is physically reasonable, as impoundment introduces more complex stress redistribution, wetting-induced deformation, and hydro-mechanical interactions, thereby increasing unresolved model discrepancy. As operating conditions gradually stabilize and additional monitoring data further constrain the effective parameter distributions, part of this discrepancy is reduced. Therefore, the inferred residual discrepancy provides a probabilistic measure of the remaining model-form and response-level uncertainty throughout the dam lifecycle.

4.5.2. *Evolution of material behavior and probabilistic settlement prediction*

Based on the sequentially updated posterior distributions, the lifecycle evolution of representative material parameters was analyzed and further propagated through the trained surrogate model for probabilistic settlement prediction. This analysis links the inferred material-state changes with the observed deformation behavior of the dam during construction, impoundment, and operation.

Figure 11 shows the posterior evolution of key constitutive parameters across the four lifecycle phases. For the main rockfill zone, the posterior mean of the stiffness parameter K_1 increases from Phase I to Phase II, reflecting progressive compaction during staged filling. It decreases in Phase III because initial impoundment induces wetting-related softening and particle rearrangement in the upstream rockfill. In Phase IV, the slight increase of K_1 should be interpreted as an equivalent structural-scale response rather than a recovery of particle-scale stiffness; it may be associated with stress redistribution, continued compaction, and re-equilibration under sustained geostatic and hydraulic loading. In contrast, the stiffness parameter K_4 of the core wall shows smaller fluctuations, indicating a more stable material response under stronger confinement and a relatively uniform stress environment.

The time-dependent and wetting-related parameters also exhibit clear stage-dependent behavior. During Phases I-II, construction-induced settlement dominates, whereas creep and wetting effects are not yet fully developed; therefore, their posterior distributions remain relatively broad. From Phase III onward, impoundment and accumulated monitoring data provide stronger information on rheological and wetting-induced deformation, leading to more concentrated posterior distributions and improved parameter identifiability. These trends are consistent with the expected mechanisms of construction compaction, impoundment-induced wetting, and long-term deformation adjustment.

Figure 12 presents the sequential probabilistic settlement predictions at two representative monitoring points, CH76 in the rockfill zone and DC4-24 in the core wall, obtained by propagating the posterior parameter samples through the trained surrogate model at each lifecycle stage. At each phase, only monitoring data available up to the current updating stage were used for calibration, while the subsequent period was treated as a forward prediction interval. In the early stages, limited monitoring information leads to wider prediction intervals. As additional data are assimilated, the posterior distributions become more constrained, the

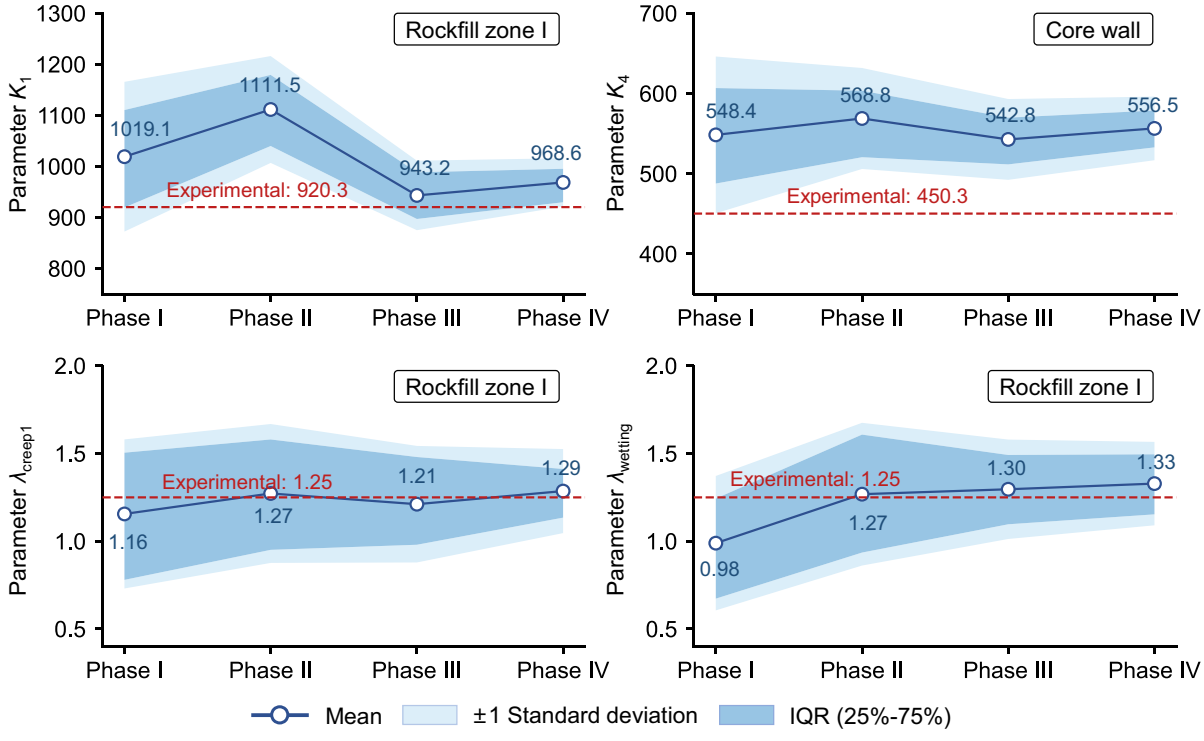


Fig. 11. Probabilistic evolution of key constitutive parameters across lifecycle phases.

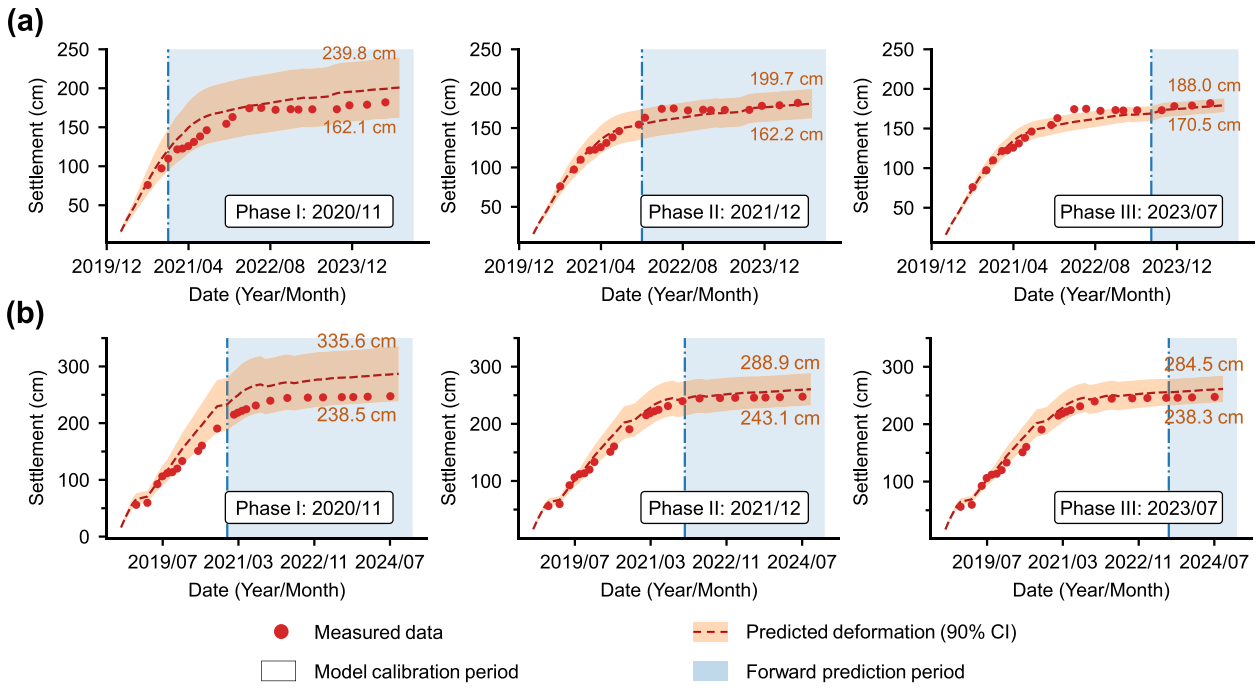


Fig. 12. Sequential probabilistic prediction of settlement at representative monitoring points: (a) CH76 in the rockfill zone; (b) DC4-24 in the core wall.

prediction intervals narrow, and the posterior mean predictions better follow the measured settlement trends. Most observations fall within the predicted credible intervals, indicating that the framework can progressively improve forward settlement prediction while quantifying uncertainty.

To further evaluate the robustness and generalization capability of the proposed framework, Fig. 13 compares the predicted settlements with the measured settlements at three representative monitoring points selected from the independent validation dataset defined in Section 4.3. This comparison provides an independent evaluation of the posterior predictive performance at locations not used for model calibration. For a fair comparison, all monitoring-based methods employed the same monitoring dataset accumulated up to the latest lifecycle stage and the same surrogate-assisted forward model. The traditional simulation used laboratory-derived parameters without monitoring-based updating. The deterministic inverse analysis followed a recently proposed physics-constrained inversion framework (Huang et al., 2026), in which the NSGA-III algorithm was used to identify a single optimal parameter set by minimizing the discrepancy between simulated and monitored settlements. The standard Bayesian updating adopted the same likelihood formulation as the proposed framework but assumed independent parameter priors. The proposed framework differs only in introducing a Gaussian Copula prior to represent the empirical dependence among parameters. The results show that the traditional simulation exhibits noticeable deviations from the measured settlements, indicating that laboratory-derived parameters cannot fully capture the in-situ mechanical behavior of the prototype dam. Although deterministic inversion improves prediction accuracy through parameter calibration, it provides only a single deterministic prediction and cannot quantify predictive uncertainty. In contrast, the proposed probabilistic framework not only yields settlement predictions that agree well with the monitoring data but also provides credible intervals for deformation responses. Its satisfactory performance at the independent monitoring point further demonstrates the effectiveness of the Bayesian updating framework and its capability to generalize to locations not directly involved in the updating process.

Overall, the proposed framework enables a physically interpretable and probabilistically consistent characterization of dam deformation throughout the lifecycle. The results demonstrate its effectiveness for reliable deformation prediction and uncertainty-informed dam safety assessment.

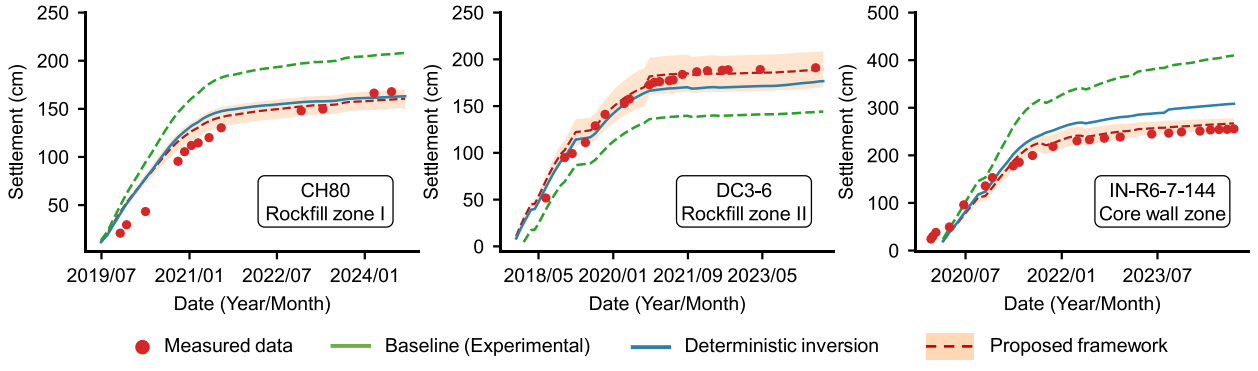


Fig. 13. Comparative evaluation of deformation predictions at independent validation points excluded from model updating.

4.5.3. Lifecycle safety assessment based on the updated deformation model

Beyond probabilistic prediction at representative monitoring locations, safety-oriented dam assessment requires characterizing the overall deformation state and identifying potential risk-prone regions throughout the dam lifecycle. In this study, settlement-related quantities of interest were evaluated probabilistically by propagating posterior parameter samples through the surrogate model, whereas full-field safety-related indicators were diagnosed using the high-fidelity FE model updated with posterior mean parameters. In this way, the proposed framework links uncertainty-informed settlement prediction with posterior-informed structural response diagnosis.

Figure 14 presents the probabilistic evolution of the maximum settlement at representative sections and within the core wall. The maximum settlement reflects the cumulative deformation level and long-term deformation tendency of the dam. As monitoring data from construction, impoundment, and operation stages were progressively assimilated, the prediction intervals gradually narrowed, indicating reduced deformation-response uncertainty and improved stability of long-term settlement prediction. The maximum settlement increased with lifecycle progression at all sections. At the final stage, the largest posterior mean settlement occurred at section 3-3, reaching 356.2 cm, followed by the core wall with 335.4 cm. This indicates that deformation accumulation was mainly concentrated in the central dam body and near the core wall.

Although the surrogate-assisted Bayesian framework enables efficient probabilistic prediction of settlement responses at key locations, comprehensive dam safety assessment requires extending the analysis from discrete response prediction to continuous full-field deformation evaluation, as safety-related indicators rely on continuous deformation and stress fields. Therefore, the posterior mean parameters at each lifecycle stage were adopted as the most probable updated material state and incorporated into the high-fidelity FE deformation model. The reconstructed full-field responses provide a posterior-informed representation of the current dam behavior and support the diagnosis of safety-related deformation characteristics.

The reconstructed settlement fields are shown in Fig. 15. At both the representative 3-3 section and the core wall section, the settlement contours exhibit a nearly concentric pattern, with the maximum settlement concentrated near the dam axis and core wall. As filling and impoundment progressed, the location of maximum settlement migrated upward from the lower-middle dam body and gradually stabilized at approximately one-half to two-thirds of the dam height. This pattern is consistent with progressive deformation driven by construction compaction, stress redistribution, reservoir impoundment, and environmental effects. The settlement fields remained smooth and continuous, and no abnormal localized discontinuities were observed.

The core-wall arching effect was further evaluated using the posterior-updated FE model. In tall core-wall rockfill dams, deformation incompatibility between the relatively compressible core wall and the adjacent rockfill zones may induce shear stress transfer along the core–rockfill interfaces, causing the actual vertical stress in the core wall to be lower than the theoretical overburden stress. Excessive arching may reduce the effective stress in the core wall and weaken its seepage stability and crack resistance, especially near dam abutments or geometrically discontinuous regions. To quantify this effect, the arching coefficient R_a was defined as the ratio of the vertical stress σ_z to the overburden stress γh_z (Handy, 1985; Sukkarak et al., 2019; Y. Zeng et al., 2018):

$$R_a = \frac{\sigma_z}{\gamma h_z} \tag{14}$$

where σ_z is the vertical stress, γ is the unit weight of the soil, and h_z is the thickness of the overlying soil column. A smaller R_a indicates a stronger arching effect in the core wall.

As shown in Fig. 16(a), the arching coefficient exhibits clear spatial heterogeneity and stage-dependent evolution. Relatively low R_a values appeared near the bottom of the core wall because local geometric discontinuity caused stress redistribution in this region. At dam completion, the deformation mismatch between the core wall and adjacent rockfill zones generated strong interface shear stresses and significant unloading of vertical stress in the core wall, leading to a pronounced arching effect. During initial impoundment, hydraulic loading and wetting-induced deformation in the upstream rockfill reduced the deformation mismatch between the core wall and rockfill. Consequently, R_a increased on the upstream side, indicating partial release of the arching effect. At full impoundment, sustained hydraulic loading and long-term rheological deformation further weakened the upstream interface constraint, and the arching effect tended toward a more stable state. In contrast, R_a on the downstream side changed only slightly, suggesting that impoundment-induced stress redistribution was mainly concentrated on the upstream side.

Because the dam crest is sensitive to differential settlement, and nonuniform deformation may induce cracking of the impervious core, the deformation gradient was adopted to evaluate potential cracking risk. This index has been widely used to assess differential settlement and crest cracking risk (Ai et al., 2025) and is calculated as:

$$\gamma = \frac{\delta_a - \delta_b}{L_{ab}} = \tan\theta \approx \theta \tag{15}$$

where δ_a and δ_b are the settlements at the two points, L_{ab} is their horizontal spacing, and θ is the inclination angle between the connecting line and the horizontal plane. In engineering practice, a threshold of $\gamma_c = 1.0\%$ is commonly adopted to assess crest cracking risk.

The crest deformation gradient calculated from the posterior-updated model is shown in Fig. 16(b). At dam completion (Phase II), the maximum gradient was only 0.32%, indicating relatively uniform settlement and good deformation compatibility. During initial impoundment (Phase III), hydraulic loading and wetting effects increased differential settlement along the crest, and the maximum gradient rose to 0.92%, still below the control threshold. At full impoundment (Phase IV), sustained high water level, wetting deformation, and creep further intensified spatial nonuniformity, increasing the maximum gradient to 1.19%. Some local regions exceeded the 1.0% threshold, suggesting potential cracking risk. High-gradient zones began to form during initial impoundment and expanded toward the central crest and the right bank at full impoundment, reflecting deformation incompatibility induced by hydraulic loading, time-dependent material behavior, and abutment constraints.

Overall, the posterior-updated deformation model captures the spatiotemporal evolution of dam deformation and provides a mechanistic basis for identifying safety-critical deformation characteristics. The results show that the proposed sequential updating framework can

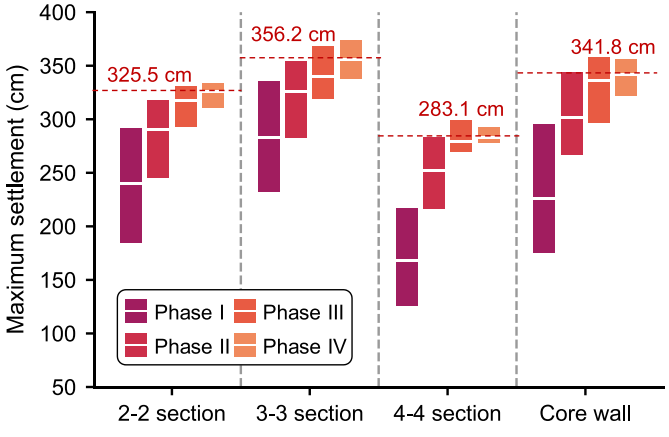


Fig. 14. Probabilistic evolution of maximum settlement across lifecycle stages.

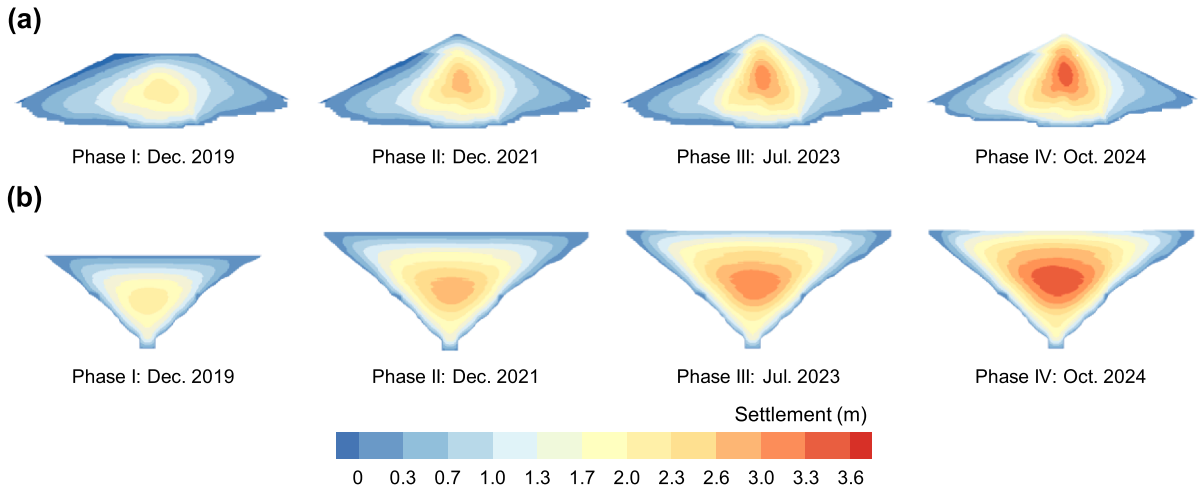


Fig. 15. Spatiotemporal evolution of settlement contours for the rockfill dam: (a) Typical cross-section 3.3; (b) Core wall section.

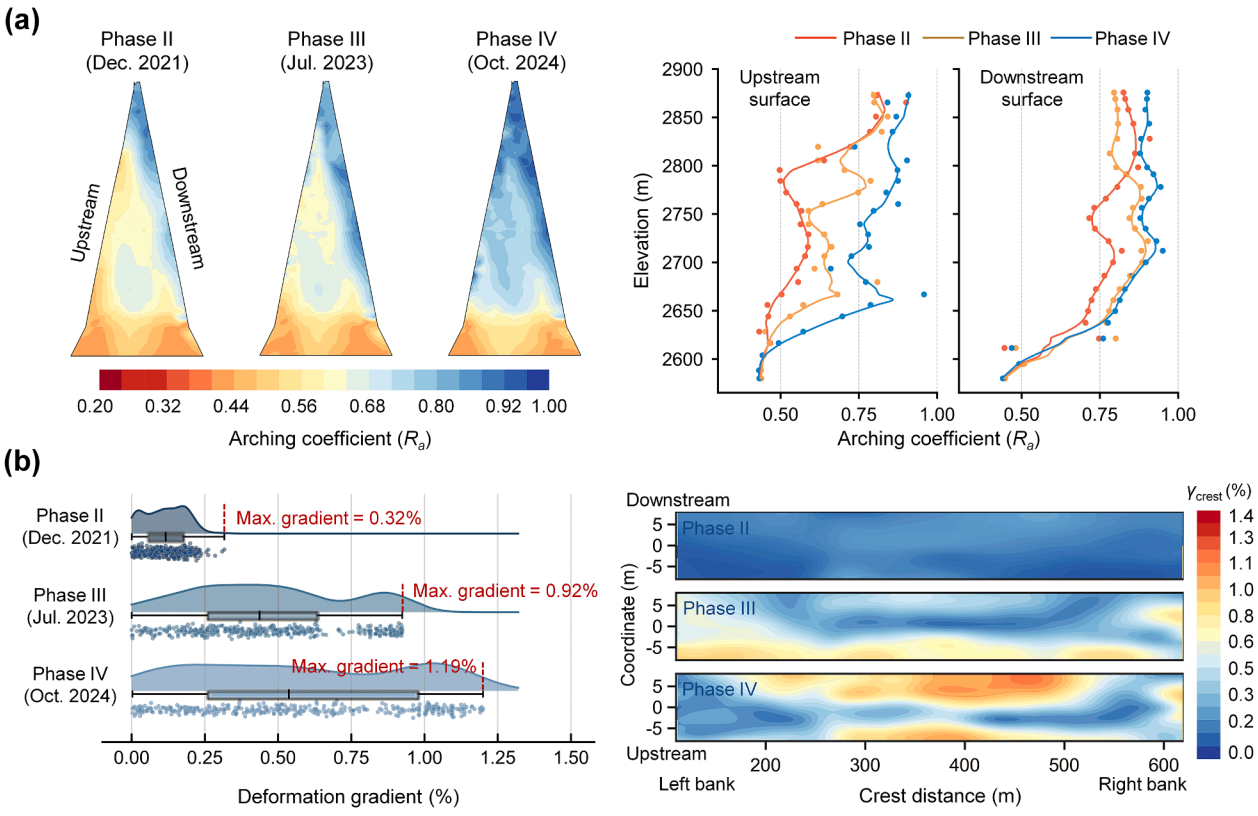


Fig. 16. Lifecycle evolution of safety-related structural responses based on the updated deformation model: (a) spatiotemporal evolution of the core-wall arching effect; (b) spatial evolution of the crest deformation gradient.

progressively improve the reliability of lifecycle deformation analysis and support posterior-informed safety diagnosis for digital twin-driven rockfill dam management.

5. Validation and discussion

5.1. Assessment of Copula-enhanced priors for Bayesian updating

Parameter dependence is an important factor affecting the interpretability and reliability of Bayesian updating in rockfill dam deformation analysis. Parameters such as K, K_b, φ, and $\Delta\varphi$ jointly describe the stiffness and strength characteristics of rockfill materials and are therefore not expected to vary independently from an engineering perspective. To represent this dependence, the proposed framework introduces a Gaussian Copula prior as a dependence-informed prior that preserves the empirical monotonic correlations among mechanically related constitutive parameters. Its purpose is not to impose exact material laws, but to reduce unrealistic decoupling caused by parameter compensation in displacement-based Bayesian updating.

Figure 17 compares the lifecycle posterior evolution and final-stage correlation structures obtained using the Gaussian Copula prior and the independent prior. As shown in Fig. 17(a), the posterior means of K,

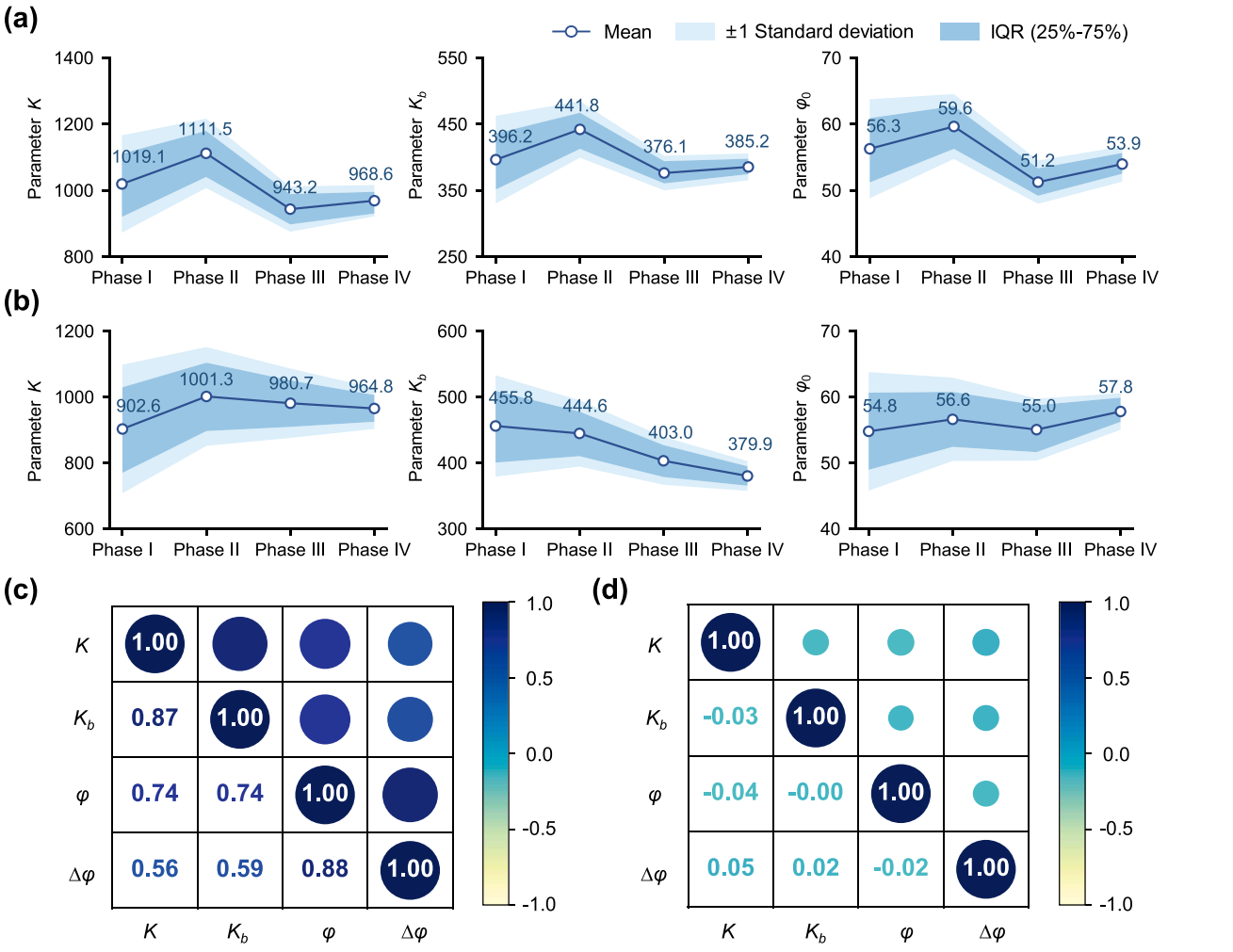


Fig. 17. Comparison of posterior parameter evolution and final-stage correlation structure: (a) Posterior evolution with Copula prior; (b) Posterior evolution with independent prior; (c) Correlation matrix at the final phase with Copula prior. (d) Correlation matrix at the final phase with independent prior.

K_b, and φ_0 under the Copula prior exhibit coordinated evolution across lifecycle stages, indicating that the prescribed empirical dependence among mechanically related parameters is largely retained. This evolution is consistent with the coupled stiffness–strength behavior of rockfill materials discussed in Section 4.5.2. In contrast, although Bayesian updating with independent priors can still achieve a reasonable fit to the monitoring observations, the posterior trajectories show much weaker dependence because different parameter combinations may produce similar settlement responses. Such compensation effects may cause originally correlated parameters to evolve independently or even in opposite directions, reducing the interpretability of the inferred parameter states.

The final-stage posterior correlation matrices further confirm this difference. With the Gaussian Copula prior, clear positive correlations are retained among stiffness- and strength-related parameters, with correlation coefficients of 0.87 between K and K_b and 0.88 between φ_0 and $\Delta\varphi_0$. By contrast, the independent-prior Bayesian updating produces correlations close to zero, indicating largely decoupled posterior samples. These results show that the Copula prior effectively preserves the prescribed empirical dependence structure and improves the interpretability of posterior parameter evolution.

To examine whether the preservation of parameter dependence also affects prediction-level behavior, Table 8 further compares the settlement prediction performance of different deformation analysis strategies using the independent validation dataset defined in Section 4.3. The global relative error was calculated by comparing the measured settlement records at the independent validation points with the corresponding model predictions from each method. For a fair comparison, all monitoring-based methods used the same monitoring dataset and the same surrogate-assisted forward model; therefore, the differences mainly reflect the updating strategy and the treatment of parameter dependence. The traditional simulation based on laboratory-derived

Table 8
Comparative assessment of global settlement prediction and parameter-dependence treatment.

Method	Global relative error (%)	Predicted maximum settlement (cm)	90% Credible interval (cm)	Treatment of parameter dependence
Traditional simulation	37.4	462.4	-	Empirical correlations
Deterministic inverse analysis	11.8	342.7	-	Ignored*
Standard Bayesian updating	10.2	367.9	[320.2, 405.6]	Ignored
Proposed framework	9.7	356.2	[338.1, 373.8]	Explicitly (Copula prior)

parameters produced the largest global relative error and clearly overestimated the maximum settlement, indicating that laboratory parameters cannot fully represent the equivalent mechanical behavior of the prototype dam. Deterministic inverse analysis substantially reduced the error, but it yielded only a single optimal parameter set and could not quantify parameter or prediction uncertainty.

Note: The global relative error was calculated from the measured settlements at the independent validation points defined in Section 4.3 and the corresponding model predictions. The observed final-stage maximum settlement was 351.6 cm. For Bayesian methods, the predicted value denotes the posterior predictive mean. *Parameter dependence is ignored in standard deterministic inversion, although it can be preserved by recent physics-constrained regularization methods (Huang et al., 2026).

The comparison between the two Bayesian methods further highlights the role of parameter-dependence treatment. Standard Bayesian updating with independent priors achieved competitive prediction accuracy through monitoring-data assimilation, but its predicted maximum settlement showed a relatively wide 90% credible interval. This wider interval is mainly attributable to the weaker dependence constraints among mechanically related parameters, which allow more compensatory parameter combinations to remain in the posterior samples. Although these combinations can produce similar settlement fits at the calibration locations, they lead to greater dispersion in posterior predictive settlement responses. In contrast, the proposed Copula-enhanced Bayesian framework achieved the lowest global relative error and predicted a final maximum settlement of 356.2 cm, close to the observed value of 351.6 cm. Its 90% credible interval was more concentrated while still covering the observed maximum settlement. These results indicate that the Copula-informed prior improves posterior parameter interpretability and enhances the stability of posterior predictive settlement responses.

5.2. Effectiveness and computational efficiency of the PG-CSL strategy

In lifecycle deformation analysis of rockfill dams, the surrogate model must adapt to newly available monitoring data and evolving boundary conditions while retaining previously learned deformation-response mappings. This section evaluates the proposed posterior-guided continual surrogate learning (PG-CSL) strategy in terms of historical knowledge retention and computational efficiency.

To assess knowledge retention, the final surrogate model obtained after full lifecycle updating was retrospectively evaluated on Phases I-III using 100 independent FE test samples. As shown in Fig. 18(a), Fine-Tune exhibits obvious performance degradation on earlier phases, indicating catastrophic forgetting caused by overwriting previously learned representations. LwF and EWC alleviate this degradation to some extent, but their prediction accuracy still decreases over long task sequences. In contrast, the proposed PG-CSL strategy maintains consistently higher accuracy across all historical phases, with only minor inter-phase variation. This indicates that combining Fisher-weighted regularization with representative historical replay effectively preserves stage-wise deformation patterns while allowing the surrogate to adapt to new lifecycle conditions.

The computational efficiency of the PG-CSL strategy was further evaluated by comparing the cumulative FE sample-generation cost with that of conventional full retraining. All FE simulations were performed on a workstation equipped with an Intel® Core™ i7-14700 processor and 32 GB RAM, and the average wall-clock time for a single high-fidelity FE simulation was approximately 5.5 h. Under the surrogate-assisted Bayesian framework, neural-network training time and NUTS-MCMC sampling time were of the same order for the compared strategies; therefore, the dominant computational burden arose from generating high-fidelity FE samples for surrogate construction or updating.

As shown in Fig. 18(b), the full-retraining baseline regenerates a new surrogate dataset and retrains the model from scratch at each lifecycle phase, requiring 1200 FE simulations in total and resulting in a cumulative FE sample-generation time of 6600 h by Phase IV. In contrast, the proposed PG-CSL strategy reuses historical surrogate knowledge and requires only posterior-guided incremental FE samples after Phase I. Specifically, it uses 300 initial samples in Phase I and 150, 100, and 100 additional samples in Phases II-IV, respectively, corresponding to 650 FE simulations and a cumulative FE sample-generation time of 3575 h. Thus, the proposed strategy saves 3025 h, corresponding to a 45.8% reduction in the dominant FE sample-generation cost. These results demonstrate that PG-CSL improves both knowledge retention and computational feasibility, making long-term sequential updating more practical for lifecycle digital-twin applications.

5.3. Limitations and future work

Although the proposed framework demonstrates good performance in lifecycle deformation updating and posterior-informed safety diagnosis, several limitations remain and provide directions for future work.

First, the finite-element discretization uncertainty has been incorporated into the likelihood through a practical quantity-of-interest-oriented estimate, which helps mitigate overconfident posterior inference caused by numerical error. Nevertheless, more rigorous goal-

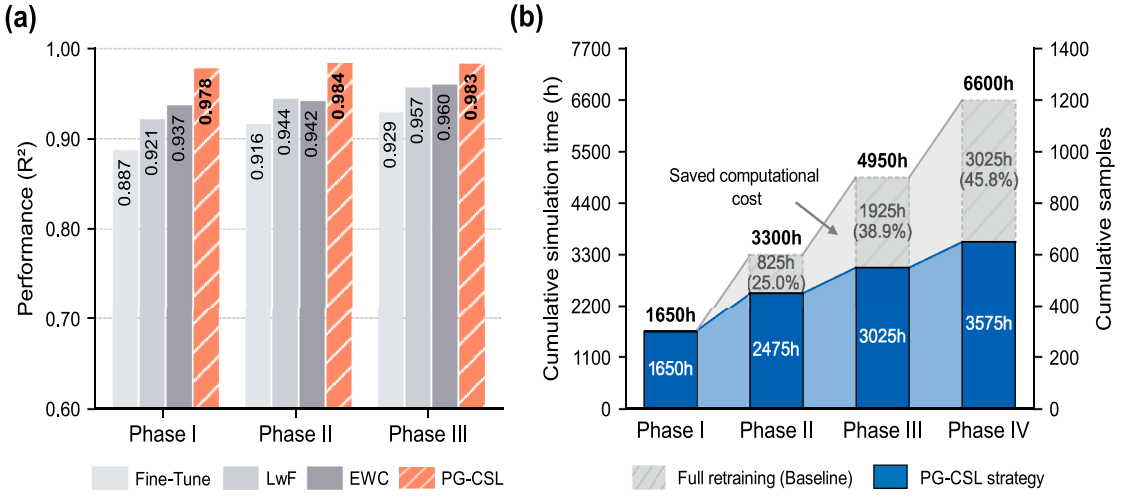


Fig. 18. Effectiveness and computational efficiency of the PG-CSL strategy: (a) comparison of historical-phase prediction performance; (b) cumulative FE-sample generation cost compared with full retraining.

oriented finite-element error estimation and a more comprehensive uncertainty budget for dense safety-related indicators remain valuable directions for further improvement. Future work may therefore integrate goal-oriented a posteriori error estimation with stochastic finite-element uncertainty propagation or derivative-accelerated Monte Carlo techniques to achieve a more systematic and reliable uncertainty characterization for safety-related quantities of interest (Duprez et al., 2020; Hauseux et al., 2017, 2018).

Second, the likelihood and sequential updating formulations can be further refined for long-term multi-source monitoring data. In this study, sensor-specific observation precision is considered, whereas possible sensor bias, environmental effects, temporal correlation, and spatially correlated residuals are not explicitly modeled. These effects are partly absorbed by the inferred residual discrepancy term, but hierarchical, bias-aware, or spatially correlated likelihood models could provide a more detailed description when denser monitoring data are available. In addition, the updated constitutive parameters are interpreted as stage-wise effective material states, which is computationally feasible for large-scale dam analysis. A full state-space Bayesian formulation with explicit parameter-transition models and process-noise terms would be more suitable for representing continuously evolving wetting, creep, and stress-redistribution effects (Rappel et al., 2018).

Third, the dependence model and probabilistic safety diagnosis can be further extended. The Gaussian Copula prior provides a parsimonious way to preserve prescribed monotonic dependence among mechanically related rockfill parameters, but it cannot fully capture tail dependence or complex nonlinear dependence. More flexible dependence models, such as *t*-Copulas, vine Copulas, or data-driven dependence representations, may be explored when larger databases or richer project-specific information become available. Moreover, the present surrogate mainly supports posterior predictive analysis of settlement-related responses, whereas full-field safety indicators, such as the core-wall arching coefficient and crest deformation gradient, are evaluated using the posterior-mean updated FE model. Future work should develop full-field surrogate models or reduced-order reconstruction techniques to estimate credible intervals and exceedance probability maps for safety-related indicators, and further integrate seepage, stress, and multi-modal monitoring data for multi-physics digital-twin assessment of rockfill dams.

6. Conclusions

This study proposes a dependence-aware sequential probabilistic framework with posterior-guided continual surrogate learning for lifecycle deformation analysis of rockfill dams. The framework combines physics-based finite-element simulation, Gaussian Copula-enhanced Bayesian updating, and continual surrogate adaptation to assimilate monitoring data, quantify parameter uncertainty, preserve prescribed parameter dependence, and reduce cross-stage surrogate updating costs. The proposed method was validated using the 303 m Lianghekou core-wall rockfill dam. The main conclusions are as follows:

(1) The proposed sequential Bayesian updating framework enables lifecycle probabilistic tracking of stage-wise effective constitutive parameters of rockfill materials. As monitoring data are progressively assimilated, the posterior distributions become more concentrated and the inferred parameter evolution reflects the effects of construction compaction, impoundment-induced wetting, stress redistribution, and long-term deformation adjustment. By embedding a Gaussian Copula dependence structure into the posterior-to-prior propagation process, the framework preserves prescribed empirical correlations among mechanically related parameters, thereby improving the physical interpretability and stability of the updated material-state identification.

(2) The PG-CSL strategy provides an efficient mechanism for cross-stage surrogate evolution. By combining posterior-guided incremental FE sampling, representative historical replay, and Fisher-weighted regularization, the surrogate model adapts to new lifecycle conditions while retaining historical deformation-response mappings. Compared with full retraining, the proposed strategy reduces the number of required FE simulations from 1200 to 650 and lowers the dominant FE sample-generation cost by 45.8%, making long-term sequential updating more computationally feasible.

(3) The proposed framework improves lifecycle deformation prediction and supports safety-oriented assessment of the updated dam state. In the Lianghekou dam case, it achieves the lowest global relative error of 9.7% on independent validation data and provides credible intervals for settlement prediction. The posterior-updated deformation analysis model further enables full-field deformation reconstruction and posterior-informed diagnosis of safety-related indicators, such as the core-wall arching effect and crest deformation gradient. These results show that the framework provides a practical basis for lifecycle deformation analysis and digital twin-driven safety management of rockfill dams.

Overall, the proposed framework provides a computationally feasible and uncertainty-informed approach for lifecycle deformation model updating of rockfill dams. Future work will further extend the framework toward full-field probabilistic safety indicators, more rigorous uncertainty-budget construction, multi-physics coupling, and richer multi-source monitoring-data assimilation.

CRediT authorship contribution statement

Zhihong Huang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Gang Ma:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Yi Zhang:** Writing – review & editing, Investigation. **Guike Zhang:** Resources, Funding acquisition. **Zhitao Ai:** Writing – review & editing, Investigation. **Yifan Cui:** Writing – review & editing, Investigation. **Wei Zhou:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Rockfill dam database and prior parameter settings

Table A.1
Summary of the rockfill dams that provide the experimental data.

No.	Dam	Type	Height(m)	Rockfill lithology	Year	Country
1	Houziyan	CFRD	223.5	Tuff and rhyolite	2018	China
2	Jinfoshan	CFRD	109.8	Tuff	2020	China
3	Jiangpinghe	CFRD	219	Tuff	2019	China
4	Nam Ou VI	CFRD	88	Slaty and sandstone	2015	Laos
5	Lianghekou	CWRD	303	Siltstone and slaty	2021	China
6	Bakun	CFRD	205	Siltstone and argillite	2007	Malaysia
7	Lawa	CFRD	239	Schist and marble	Constructing	China
8	Shanmichong	CFRD	104	Sandstone and slaty	Planning	China
9	Dadai	CFRD	115	Sandstone	2013	Cambodia
10	Jishixia	CFRD	103	Sandstone	2011	China
11	Jinchuan	CFRD	112	Sandstone	Constructing	China
12	Dashixia	CFRD	247	Sandstone	Constructing	China
13	Guanfeng	CFRD	104.7	Rhyolite	Constructing	China
14	Fangxian	CFRD	105	Quartzite and granulite	Planning	China
15	Yele	CWRD	124.5	Quartzite	2005	China
16	Maerdang	CFRD	211	Monzonite and sandstone	2023	China
17	Pushihe	CFRD	78.5	Limestone	2011	China
18	Jiayan	CFRD	154	Limestone	2021	China
19	Zipingpu	CFRD	156	Limestone	2006	China
20	Tianshengqiao	CFRD	178	Limestone	2000	China
21	Shuibuya	CFRD	233	Limestone	2007	China
22	Turimiquire	CFRD	115	Limestone	1980	Venezuela
23	Laodukou	CFRD	96.8	Gravels	2009	China
24	Cihaxia	CFRD	257.5	Gravels	Constructing	China
25	Lianhua	CFRD	71.8	Granite	1996	China
26	Shahe	CFRD	77	Granite	2024	China
27	Gongboxia	CFRD	132.2	Granite	2002	China
28	Maji	CFRD	277.5	Gneiss	Unrealized	China
29	Wawushan	CFRD	138	Dolerite and sandstone	2007	China
30	Baini	CFRD	52	Dolerite	Constructing	China
31	Rumei	CWRD	315	Dacite	Constructing	China
32	Langhe	CWRD	46.2	Clay and Gravels	1971	China
33	Sifangjing	CWRD	51.3	Clay and gravels	2021	China
34	Nuozhadu	CWRD	261.5	Clay and gravels	2012	China
35	Shuangjiangkou	CWRD	315	Clay and gravels	Constructing	China
36	Tacheng	CWRD	315/208	Clay and gravels	Planning	China
37	Niuniu	CFRD	155	Basalt and limestone	2004	China
38	Tianxingba	CFRD	87	Basalt	2023	China
39	Quxue	CWRD	164.2	Basalt	2016	China
40	Gushui	CWRD	305	Basalt	Planning	China
41	Hanjiaxia	CFRD	38.6	-	2023	China
42	Xianyou	CFRD	72.4	-	2013	China
43	Xilongchi	CFRD	97	-	2008	China
44	Jinzhong	CFRD	97.5	-	2010	China
45	Guwa	CFRD	139	-	2021	China
46	Kárahnjúkar	CFRD	193	-	2008	Iceland
47	Baoquan	CWRD	94.8	-	2007	China
48	Comedero	CWRD	136	-	1991	Mexico

Note: CWRD denotes core wall rockfill dam; CFRD denotes concrete-faced rockfill dam.

Table A.2
Initial constitutive model parameters for different material zones based on laboratory experiments.

Constitutive model	Parameter	Rockfill zone I	Rockfill zone II	Rockfill zone III	Core wall
Duncan-Chang E-B model	K	920	853	783	450
	K_b	290	280	287	350
	n	0.29	0.25	0.27	0.41
	m	0.20	0.16	0.19	0.39
	φ_0	49.2	49.2	46.8	23.0
	$\Delta\varphi$	5.0	8.1	5.3	-
	c (kPa)	183	260	197	122
	R_f	0.75	0.75	0.75	0.88
Exponential decay empirical creep model	α	0.0007	0.0007	0.0007	0.0007
	b	0.0194	0.0194	0.0194	0.0194
	c	0.0056	0.0056	0.0056	0.0056
	d	0.0296	0.0296	0.0296	0.0296
Three-parameter wetting deformation model	c_w	0.0006	0.0006	0.0006	-
	n_w	0.620	0.620	0.620	-
	b_w	0.003	0.003	0.003	-

Table A.3
Prior ranges of constitutive model parameters for surrogate training.

Constitutive model	Parameter	Rockfill zone I	Rockfill zone II	Rockfill zone III	Core wall
Duncan-Chang E-B model	K	600-1200	550-1150	500-1100	200-750
	K_b	200-500	200-500	200-500	200-700
	n	0.2-0.5	0.2-0.5	0.2-0.5	0.2-0.5
	m	0.15-0.50	0.15-0.50	0.15-0.50	0.15-0.50
	φ_0	32-64	32-64	32-64	15-32
	$\Delta\varphi$	5-12	5-12	5-12	-
Exponential decay empirical creep model	α ($\times 10^{-3}$)	0.5-1.5	0.5-1.5	0.5-1.5	0.5-1.5
	λ_{creep}	0.5-2.0	0.5-2.0	0.5-2.0	0.5-2.0
Three-parameter wetting deformation model	λ_{wetting}	0.5-2.0	0.5-2.0	0.5-2.0	-

Appendix B. NUTS-MCMC convergence diagnostics

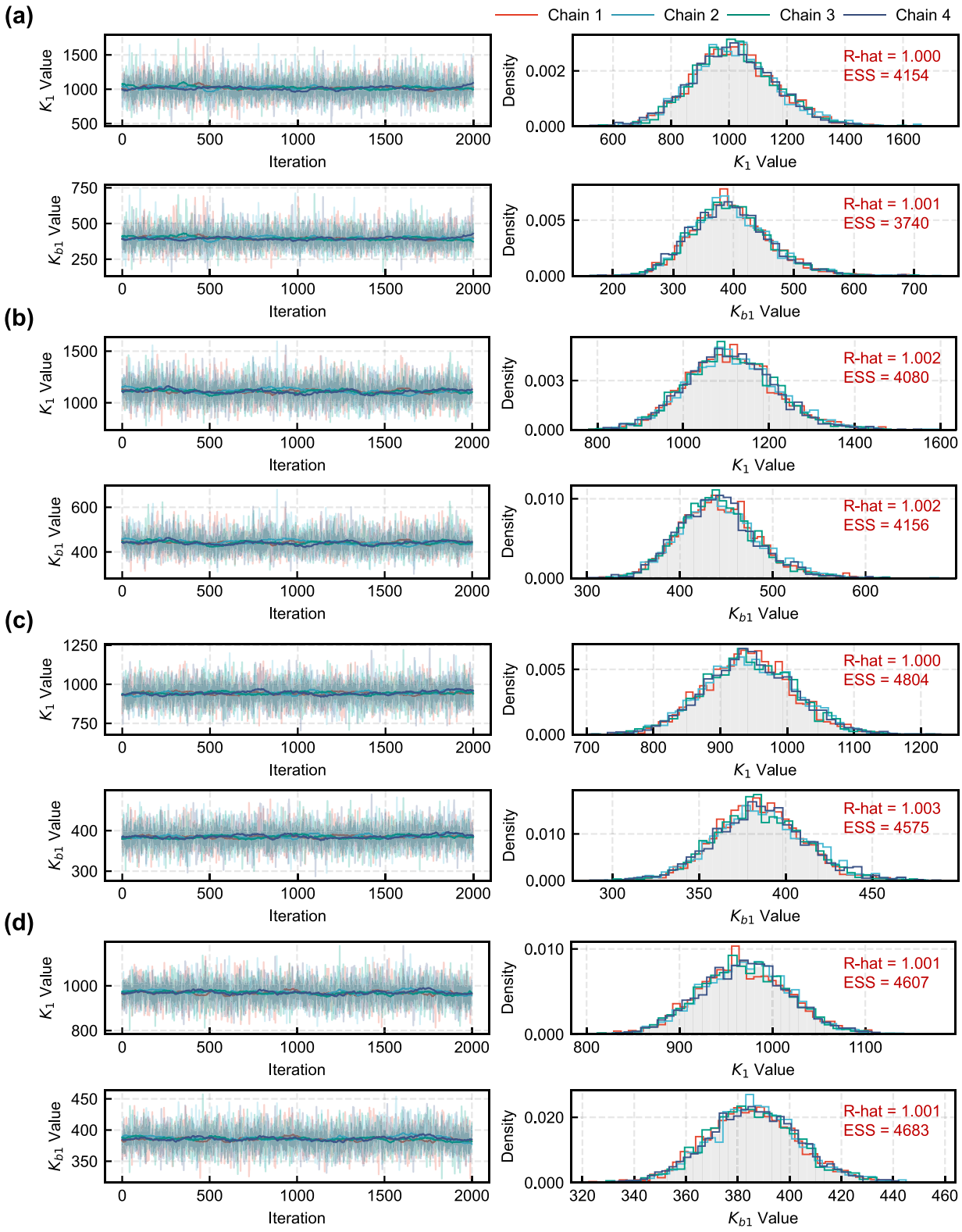


Fig. B.1. MCMC trace plots and posterior density diagnostics. (a) Phase I; (b) Phase II; (c) Phase III; (d) Phase IV.

Data availability

Data will be made available on request.

References

Ai, Z., Ma, G., Qu, T., Zhang, G., Cui, Y., & Zhou, W. (2026). Generative digital twin for deformation field reconstruction of high rockfill dams based on diffusion models.

Computer-Aided Civil and Infrastructure Engineering, 46, Article 100090. https://doi.org/10.1016/j.cacaie.2026.100090

Ai, Z., Ma, G., Zhang, G., Liu, R., Deng, S., Chang, X., & Zhou, W. (2024). Multi-source monitoring data filtering assisted deformation analysis model updating of ultra-high rockfill dam. *Computers and Geotechnics, 171*, Article 106323. https://doi.org/10.1016/j.compgeo.2024.106323

Ai, Z., Ma, G., Zhang, G., Wang, J., Huang, Z., Zhou, W., & Yang, Q. (2025). Supervised parameter updating of deformation analyses for rockfill dams using prior knowledge. *Computer-Aided Civil and Infrastructure Engineering*. https://doi.org/10.1111/mice.70070. mice.70070.

Celik, I. B., Ghia, U., Roache, P. J., & Freitas, C. J. (2008). Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering-Transactions of the ASME, 130*(7). https://doi.org/10.1115/1.2960953

Chen, X., Sun, W., Hu, S., Li, L., Gu, C., Guo, J., Wei, B., & Xu, B. (2025). Uncertainty-informed regional deformation diagnosis of arch dams. *Computer-Aided Civil and Infrastructure Engineering, 40*(10), 1344~1369. https://doi.org/10.1111/mice.13395

Cheng, J., Ma, G., Zhang, G., Wang, Q., & Zhou, W. (2023). Deterioration of mechanical properties of rockfill materials subjected to cyclic wetting–drying and wetting. *Rock Mechanics and Rock Engineering, 56*(4), 2633~2647. https://doi.org/10.1007/s00603-022-03209-8

Coletti, K., Romeo, R. C., & Davis, R. B. (2024). Bayesian backcalculation of pavement properties using parallel transitional Markov chain Monte Carlo. *Computer-Aided Civil and Infrastructure Engineering, 39*(13), 1911~1927. https://doi.org/10.1111/mice.13123

Dai, W., Dai, Y., Xie, J., Shen, S., Shen, G., & Wang, Y. (2025). Enhancing Bayesian probabilistic back-analysis efficiency using multi-type surface and subsurface monitoring data: Case study of the Baihetan left bank slope. *Computers and Geotechnics, 182*, Article 107174. https://doi.org/10.1016/j.compgeo.2025.107174

Delange, M., Aljundi, R., Masana, M., Parisot, S., Jia, X., Leonardis, A., Slabaugh, G., & Tuytelaars, T. (2021). A continual learning survey: Defying forgetting in classification tasks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11. https://doi.org/10.1109/TPAMI.2021.3057446

Deshpande, S., Lengiewicz, J., & Bordas, S. P. A. (2022). Probabilistic deep learning for real-time large deformation simulations. *Computer Methods in Applied Mechanics and Engineering, 398*, Article 115307. https://doi.org/10.1016/j.cma.2022.115307

Ding, S., Pan, J., Wang, Y., Wang, L., Xu, H., Lu, Y., & Zhao, X. (2025). Data-driven iterative calibration method for prior knowledge of earth-rockfill dam wetting model parameters. *Journal of Rock Mechanics and Geotechnical Engineering*. , Article S1674775525003233. https://doi.org/10.1016/j.jrmge.2025.04.040

Duprez, M., Bordas, S. P. A., Bucki, M., Bui, H. P., Chouly, F., Lleras, V., Lobos, C., Lozinski, A., Rohan, P.-Y., & Tomar, S. (2020). Quantifying discretization errors for soft tissue simulation in computer assisted surgery: A preliminary study. *Applied Mathematical Modelling, 77*, 709~723. https://doi.org/10.1016/j.apm.2019.07.055

Handy, R. L. (1985). The arch in soil arching. *Journal of Geotechnical Engineering, 111*(3), 302~318. https://doi.org/10.1061/(ASCE)0733-9410(1985)111:3(302)

Hauseux, P., Hale, J. S., & Bordas, S. P. A. (2017). Accelerating Monte Carlo estimation with derivatives of high-level finite element models. *Computer Methods in Applied Mechanics and Engineering, 318*, 917936. https://doi.org/10.1016/j.cma.2017.01.041

Hauseux, P., Hale, J. S., Cotin, S., & Bordas, S. P. A (2018). Quantifying the uncertainty in a hyperelastic soft tissue model with stochastic parameters. *Applied Mathematical Modelling, 62*, 86102. https://doi.org/10.1016/j.apm.2018.04.021

Hoffman, M. D., & Gelman, A. (2014). The No-U-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research, 15*, 15931623. WOS:000338420000013.

Huang, Z., Ma, G., Ai, Z., Wang, J., Chang, X., & Zhou, W. (2026). Physics-constrained digital twin framework for deformation analysis and safety assessment of high earth-rock dams. *Reliability Engineering & System Safety, 273*, Article 112348. https://doi.org/10.1016/j.ress.2026.112348

Jia, Y., & Chi, S. (2015). Back-analysis of soil parameters of the Malutang II concrete face rockfill dam using parallel mutation particle swarm optimization. *Computers and Geotechnics, 65*, 8796. https://doi.org/10.1016/j.compgeo.2014.11.013

Jia, Y., Xu, B., Chi, S., Xiang, B., Xiao, D., & Zhou, Y. (2018). Joint back analysis of the creep deformation and wetting deformation parameters of soil used in the Guanyinyan composite dam. *Computers and Geotechnics, 96*, 167177. https://doi.org/10.1016/j.compgeo.2017.10.018

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences, 114*(13), 35213526. https://doi.org/10.1073/pnas.1611835114

Li, D.-Q., Zang, H.-H., Tang, X.-S., & Rong, G. (2024). Efficient Bayesian updating for deformation prediction of high rock slopes induced by excavation with monitoring data. *Engineering Geology, 342*, Article 107772. https://doi.org/10.1016/j.enggeo.2024.107772

Li, G., Mi, Z., Fu, H., & Fang, W. (2004). Experimental studies on rheological behaviors for rockfills in concrete faced rockfill dam. *Rock and Soil Mechanics, 25*(11), 17121716. https://doi.org/10.16285/j.rsm.2004.11.007

Li, Q., Ni, P., Du, X., & Han, Q. (2025). Bayesian model updating with variational inference and Gaussian copula model. *Computer Methods in Applied Mechanics and Engineering, 438*, Article 117842. https://doi.org/10.1016/j.cma.2025.117842

Li, Y., Li, S., Ding, Z., & Tu, X. (2013). The sensitivity analysis of Duncan-Chang EB model parameters based on the orthogonal test method. *Journal of Hydraulic Engineering, 44*(7), 873879. https://doi.org/10.13243/j.cnki.slxb.2013.07.012

Li, Z., Wu, Z., Chen, J., Li, Y., Huang, H., Lu, Y., Lu, X., & Li, J. (2024). Inverse analysis of deformation moduli for high arch dams using the displacement reconstruction technique and multi-objective optimization. *Computer-Aided Civil and Infrastructure Engineering, 39*(10), 15071529. https://doi.org/10.1111/mice.13141

Ma, G., Ai, Z., Guo, C., Li, S., Chen, H., & Zhou, W. (2024). Review on deformation monitoring for high earth-rock dams. *Journal of Hydraulic Engineering, 55*(10), 11741186. https://doi.org/10.13243/j.cnki.slxb.20240361

Ma, H., & Chi, F. (2016a). Major technologies for safe construction of high earth-rockfill dams. *Engineering, 2*(4), 498~509. https://doi.org/10.1016/j.eng.2016.04.001

Ma, H., & Chi, F. (2016b). Technical progress on researches for the safety of high concrete-faced rockfill dams. *Engineering, 2*(3), 332~339. https://doi.org/10.1016/J.ENG.2016.03.010

Miao, M., Ma, G., Ai, Z., Cheng, X., Wang, J., & Zhou, W. (2025). Comparative study of inversion strategies for constitutive modelparameters of materials in high core wall rockfill dams. *Journal of Hydroelectric Engineering, 44*(1), 64~76. https://doi.org/10.11660/slfdxb.20250106

Nakamura, K., Umehara, Y., Nakahara, M., & Imai, R. (2025). 4D spatial–temporal information infrastructure for digital twin environments using Spatial IDs. *Computer-Aided Civil and Infrastructure Engineering, 40*(30), 6184~6200. https://doi.org/10.1111/mice.70146

Nav, F. M., Mirfakhar, S. F., & Snaiki, R. (2025). A hybrid machine learning framework for wind pressure prediction on buildings with constrained sensor networks. *Computer-Aided Civil and Infrastructure Engineering, 40*(19), 2816~2832. https://doi.org/10.1111/mice.13488

Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., & Wermter, S. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks, 113*, 54~71. https://doi.org/10.1016/j.neunet.2019.01.012

Rappel, H., Beex, L. A. A., & Bordas, S. P. A (2018). Bayesian inference to identify parameters in viscoelasticity. *Mechanics of Time-Dependent Materials, 22*(2), 221~258. https://doi.org/10.1007/s11043-017-9361-0

Rappel, H., Beex, L. A. A., Hale, J. S., Noels, L., & Bordas, S. P. A. (2020). A tutorial on Bayesian inference to identify material parameters in solid mechanics. *Archives of Computational Methods in Engineering, 27*(2), 361~385. https://doi.org/10.1007/s11831-018-09311-x

Rappel, H., Beex, L. A. A., Noels, L., & Bordas, S. P. A (2019). Identifying elastoplastic parameters with Bayes' theorem considering output error, input error and model uncertainty. *Probabilistic Engineering Mechanics, 55*, 28~41. https://doi.org/10.1016/j.probengmech.2018.08.004

Ren, Q., Ke, Q., He, Y., Li, M., Xiao, L., & Li, H. (2025). Parameter inverse analysis of high rockfill dams considering material uncertainty based on the EJaya-SESM model. *Advanced Engineering Informatics, 65*, Article 103306. https://doi.org/10.1016/j.aei.2025.103306

Saif, W., RazaviAlavi, S., & Kassem, M. (2024). Construction digital twin: A taxonomy and analysis of the application-technology-data triad. *Automation in Construction, 167*, Article 105715. https://doi.org/10.1016/j.autcon.2024.105715

Sukkarak, R., Jongpradist, P., & Pramthawee, P. (2019). A modified valley shape factor for the estimation of rockfill dam settlement. *Computers and Geotechnics, 108*, 244~256. https://doi.org/10.1016/j.compgeo.2019.01.001

Sun, M., Ma, G., Qu, T., Guan, S., Mei, J., Wang, J., & Zhou, W. (2026). PE-MPINN: A parameters-enhanced multiphysics-informed neural network for data assimilation of seepage-consolidation coupling problems in spatially variable soils. *Advanced Engineering Informatics, 71*, Article 104376. https://doi.org/10.1016/j.aei.2026.104376

Talasila, P., Tcherniak, D., Jensen, A. M. D., Mahato, S., Medom, J. V., Ulriksen, M. D., Abbiati, G., Schörghofer-Queiroz, A., Larsen, P. G., & Damkilde, L. (2026). A digital twin platform for structural health monitoring. *Computer-Aided Civil and Infrastructure Engineering, 50*, Article 100086. https://doi.org/10.1016/j.cacaie.2026.100086

Tian, J., Yu, R., Chen, J., Chen, C., Li, Y., Sun, X., & Huang, H. (2025). Digital twin model for analyzing deformation and seepage in high earth-rock dams. *Automation in Construction, 173*, Article 106079. https://doi.org/10.1016/j.autcon.2025.106079

Tomassini, E., García-Macías, E., & Ubertini, F. (2025). Model-based transfer learning for real-time damage assessment of bridge networks. *Automation in Construction, 180*, Article 106581. https://doi.org/10.1016/j.autcon.2025.106581

Van De Ven, G. M., Tuytelaars, T., & Tolias, A. S. (2022). Three types of incremental learning. *Nature Machine Intelligence, 4*(12), 1185~1197. https://doi.org/10.1038/s42256-022-00568-3

Wang, K., Tang, H., Wang, R., & Zhang, J.-M. (2024). Development and evaluation of a practical nonlinear elastic constitutive model for rockfill dam deformation simulation based on monitoring results. *Acta Geotechnica, 19*(6), 3467~3485. https://doi.org/10.1007/s11440-023-02098-7

Wang, X., Pan, Y., & Chen, J. (2025). Digital twin with uncertainty-informed deep learning for prompt quantitative risk assessment of deep excavation. *Computer-Aided Civil and Infrastructure Engineering, 40*(25), 4226~4252. https://doi.org/10.1111/mice.70049

Yang, F., Liu, H., Xiao, Y., Shi, J., Peng, Y., Ding, X., & Yang, X. (2025). Mechanical degradation and particle breakage of sandstone rockfill material under wetting–drying cycles. *Acta Geotechnica, 20*(5), 2237~2255. https://doi.org/10.1007/s11440-025-02545-7

Yang, H., Chen, C., Zhao, W., Lu, X., Zhang, P., & Xue, J. (2025). A novel reliability method for assessing dam slope stability by incorporating intrinsic correlations of rockfill materials. *Reliability Engineering & System Safety, 259*, Article 110961. https://doi.org/10.1016/j.ress.2025.110961

Yu, J. A., Sun, Y., Shen, Z., & Gan, L. (2025). Joint inverse analysis for instantaneous and creep deformation parameters of high rockfill dams based on surrogate models and improved multi-objective intelligent algorithm. *Expert Systems with Applications, 277*, Article 127225. https://doi.org/10.1016/j.eswa.2025.127225

Zeng, P., Feng, B., Dai, K., Li, T., Fan, X., & Sun, X. (2024). Can satellite InSAR innovate the way of large landslide early warning? *Engineering Geology, 342*, Article 107771. https://doi.org/10.1016/j.enggeo.2024.107771

Zeng, Y., Shen, Z., Gan, L., & Li, G. (2018). Influence from filter layer on arching effect of high rockfill dam with core wall. *Water Resources and Hydropower Engineering, 49*(1), 90~95. https://doi.org/10.13928/j.cnki.wrahe.2018.01.013

Zhang, L., Wu, F., Zheng, Y., Chen, L., Zhang, J., & Li, X. (2018). Probabilistic calibration of a coupled hydro-mechanical slope stability model with integration of multiple observations. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards, 12*(3), 169~182. https://doi.org/10.1080/17499518.2018.1440317

Zhang, Y., Ma, G., Qu, T., Wang, Z., Xiong, K., & Zhou, W. (2026). MoE-D2AN: A mixture-of-experts surrogate and dual-branch data assimilation network for spatiotemporal dam deformation modeling. *Advanced Engineering Informatics, 69*, Article 104122. https://doi.org/10.1016/j.aei.2025.104122

Zhou, T., Wei, Y., Jie, Y., & Zhang, Y. (2025). A novel Bayesian updating finite element model using enhanced unscented Kalman filter for time-dependent estimation of rockfill dam structure deformation. *Engineering Structures, 323*, Article 119231. https://doi.org/10.1016/j.engstruct.2024.119231

Zhou, W., Hua, J., Chang, X., & Zhou, C. (2011). Settlement analysis of the Shuibuya concrete-face rockfill dam. *Computers and Geotechnics, 38*(2), 269~280. https://doi.org/10.1016/j.compgeo.2010.10.004